

Task on Docker -03

1. Create an image from a running container.

Lets check if we have any images or not

docker images

```
[root@ip-172-31-0-135:~# docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	In Use
926046660167.dkr.ecr.us-east-1.amazonaws.com/amazon-nginx:v1	ce95c001bce2	584MB	195MB	U
amazonlinux:latest	ad882807c6c8	227MB	59.3MB	
javed:v1	ce95c001bce2	584MB	195MB	U
javeebms7/techie:v1	ce95c001bce2	584MB	195MB	U
jenkins/jenkins:lts	01c992ffef29	799MB	289MB	U
nginx:latest	5aca99593157	241MB	66MB	U
sonarqube:latest	35bedac3f40c	2.52GB	1.08GB	
tomcat:latest	247650787b97	589MB	158MB	U
ubuntu:latest	f3d28607ddd7	160MB	45.3MB	

```
root@ip-172-31-0-135:~#
```

We have running container for that ne need to start an image

docker ps

```
[root@ip-172-31-0-135:~# docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	In Use
926046660167.dkr.ecr.us-east-1.amazonaws.com/amazon-nginx:v1	ce95c001bce2	584MB	195MB	U
amazonlinux:latest	ad882807c6c8	227MB	59.3MB	
javed:v1	ce95c001bce2	584MB	195MB	U
javeebms7/techie:v1	ce95c001bce2	584MB	195MB	U
jenkins/jenkins:lts	01c992ffef29	799MB	289MB	U
nginx:latest	5aca99593157	241MB	66MB	U
sonarqube:latest	35bedac3f40c	2.52GB	1.08GB	
tomcat:latest	247650787b97	589MB	158MB	U
ubuntu:latest	f3d28607ddd7	160MB	45.3MB	

```
[root@ip-172-31-0-135:~# docker run -itd -p 80:80 nginx:latest
ccef05956a195c9235b5b7856617fdec6b76dc513af15260d1e37ec87848276a
[root@ip-172-31-0-135:~# docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS
ccef05956a19	nginx:latest	"/docker-entrypoint..."	4 seconds ago	Up 4 seco
nds	0.0.0.0:80->80/tcp, [::]:80->80/tcp	jolly_elgamal		

```
root@ip-172-31-0-135:~#
```

Lets create an image using the running container using command

docker commit <container_id> <image_name>:<tag>

and to check if the images has been created or not use command

docker images

```
Downloads — root@ip-172-31-0-135: ~ — ssh -i jenkins.pem ubuntu@ec2...

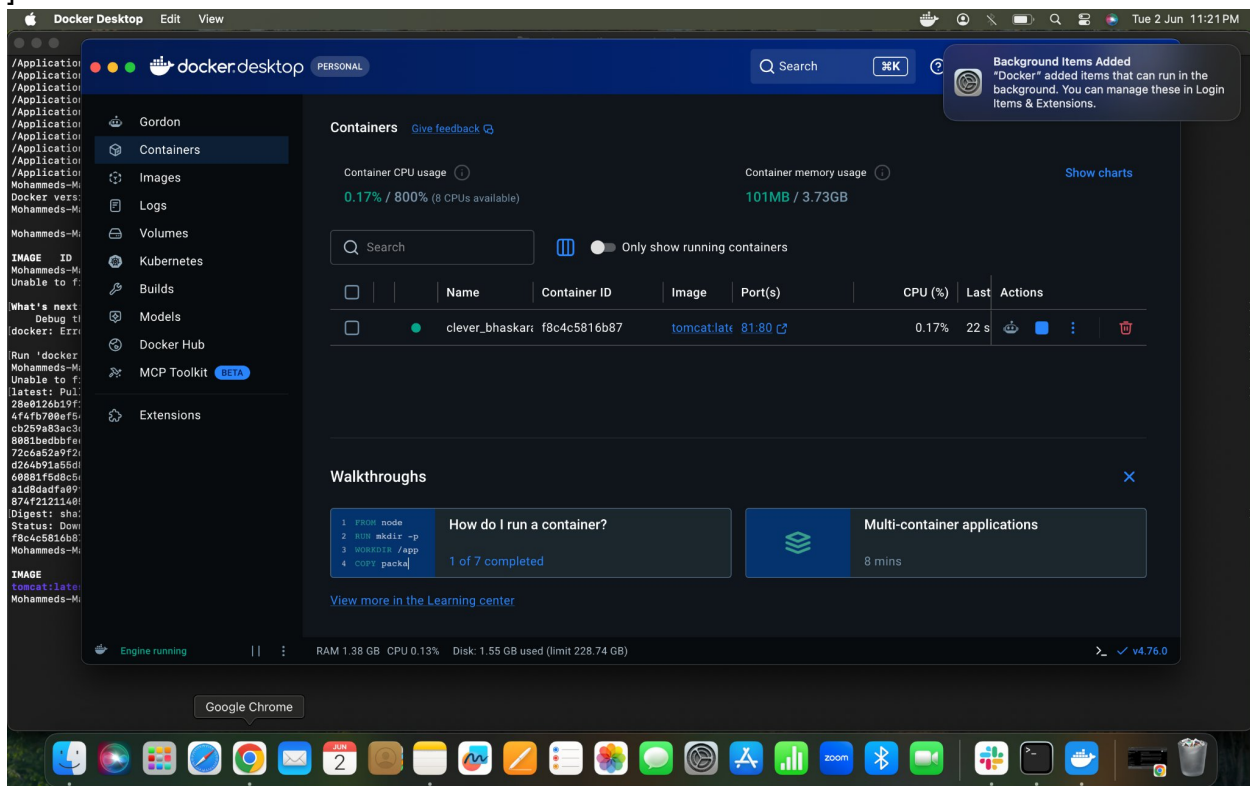
[root@ip-172-31-0-135:~# docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS
PORTS         NAMES
ccef05956a19   nginx:latest   "/docker-entrypoint..." 4 minutes ago  Up 4 minu
tes          0.0.0.0:80->80/tcp, [::]:80->80/tcp   jolly_elgamal
[root@ip-172-31-0-135:~# docker commit ccef05956a19 nginx:v1
sha256:9c845c3a54f008fa6cc1d49721b0df4b5a39304c145d0ede038b55d9c762e41d
[root@ip-172-31-0-135:~# docker images

IMAGE                                ID                                DISK USAGE  CONTENT SIZE  In Use
926046660167.dkr.ecr.us-east-1.amazonaws.com/amazon-nginx:v1    ce95c001bce2                584MB        195MB        U
amazonlinux:latest            ad882807c6c8                227MB        59.3MB
javed:v1                      ce95c001bce2                584MB        195MB        U
javeebms7/techie:v1          ce95c001bce2                584MB        195MB        U
jenkins/jenkins:lts          01c992ffef29                799MB        289MB        U
nginx:latest                  5aca99593157                241MB        66MB         U
nginx:v1                      9c845c3a54f0                238MB        63.1MB
sonarqube:latest              35bedac3f40c                2.52GB       1.08GB
tomcat:latest                 247650787b97                589MB        158MB        U
ubuntu:latest                 f3d28607ddd7                160MB        45.3MB

root@ip-172-31-0-135:~#
```

2. Copy image from local machine to docker server and load the image.

For this first Install DOcker Desktop in our local machine



Now check if docker has been installed or not by using
docker --version

```
/Applications/Docker.app/Contents/MacOS/Docker Desktop.app/Contents/Frameworks/Electron Framework
Mohammeds-MacBook-Pro:~ mohammedimran$ docker --version
Docker version 29.5.2, build 79eb04c
Mohammeds-MacBook-Pro:~ mohammedimran$ clear

Mohammeds-MacBook-Pro:~ mohammedimran$ docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
				Info → In Use

Now lets pull an tomcat image on my local machine

Docker pull tomcat

```
run docker run --help for more information
Mohammeds-MacBook-Pro:~ mohammedimran$ docker run -itd -p 81:80 tomcat:latest
Unable to find image 'tomcat:latest' locally
[latest: Pulling from library/tomcat
28e0126b19f1: Pull complete
4f4fb700ef54: Pull complete
cb259a83ac3d: Pull complete
8081bedbbfee: Pull complete
72c6a52a9f2d: Pull complete
d264b91a55d8: Pull complete
60881f5d8c56: Pull complete
a1d8dadfa09f: Download complete
874f21211405: Download complete
Digest: sha256:12dad7111b0ecf61cb493a8d4cc65d6181e4c47f1699283e7b4910b312f55736
Status: Downloaded newer image for tomcat:latest
f8c4c5816b8714679d21335859c93432ceae7be9dd1a153a9508c931bd7d528d
Mohammeds-MacBook-Pro:~ mohammedimran$ docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
tomcat:latest	12dad7111b0e	589MB	158MB	Info → In Use

As we cannot copy this image to our server we will be creating a tar file using command

Docker save -o tomcat.tar tomcat:latest

```
f8c4c5816b87 tomcat:latest "catalina.sh run" 4 minutes ago Up 4 minutes 0.0.0.0:81->80/tcp, [::]:81->80/tcp clever_bhaskara
Mohammeds-MacBook-Pro:~ mohammedimran$ docker save -o tomcat.tar tomcat:latest
Mohammeds-MacBook-Pro:~ mohammedimran$ ls
Applications Desktop Downloads key.pem Library Music Public
AWSCLI2.pkg Documents eclipse-workspace key1.pem Movies Pictures tomcat.tar
Mohammeds-MacBook-Pro:~ mohammedimran$
```

As we can see from above tar file is created now lets move the tar file to our server using command

Scp -i /Users/mohammedimran/Downloads/jenkins.pem tomcat.tar ubuntu@23.20.81.233:/tmp

```
Mohammeds-MacBook-Pro:Downloads mohammedimran$ cd ..
Mohammeds-MacBook-Pro:~ mohammedimran$ Scp -i /Users/mohammedimran/Downloads/jenkins.pem tomcat.tar ubuntu@23.20.81.233:/tmp
The authenticity of host '23.20.81.233 (23.20.81.233)' can't be established.
ED25519 key fingerprint is SHA256:5GUnxs4g9H3CBoYrVP9anR40DTK1/RRpLdhJL8qMKQ.
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:81: ec2-3-85-83-99.compute-1.amazonaws.com
  ~/.ssh/known_hosts:84: ec2-23-20-81-233.compute-1.amazonaws.com
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '23.20.81.233' (ED25519) to the list of known hosts.
tomcat.tar
Mohammeds-MacBook-Pro:~ mohammedimran$
```

Background Items Added
"Docker" added items that can run in the background. You can manage these in Login Items & Extensions.

100% 151MB 6.9MB/s 00:21

Copied successfully. Now lets check in our server if the file is existing or not

Cd /tmp ⇒ checking the tmp folder has we have given this path

```
[root@ip-172-31-0-135:~# cd /tmp/
[root@ip-172-31-0-135:/tmp# ls
snap-private-tmp
systemd-private-10c23a2ebbc54005844891fd778d2f6a-ModemManager.service-y0D0A1
systemd-private-10c23a2ebbc54005844891fd778d2f6a-chrony.service-eWRpcR
systemd-private-10c23a2ebbc54005844891fd778d2f6a-irqbalance.service-YMxhV0
systemd-private-10c23a2ebbc54005844891fd778d2f6a-polkit.service-6rF7q1
systemd-private-10c23a2ebbc54005844891fd778d2f6a-systemd-logind.service-aryHb3
tomcat.tar
root@ip-172-31-0-135:/tmp#
```

Now lets untar the file using command

Docker load -i tomcat.tar

```
tomcat.tar
[root@ip-172-31-0-135:/tmp# docker load -i tomcat.tar
Loaded image: tomcat:latest
root@ip-172-31-0-135:/tmp#
```

Now lets check in

docker images


```

root@ip-172-31-0-135:/tmp# docker load -i tomcat.tar
Loaded image: tomcat:latest
root@ip-172-31-0-135:/tmp# docker images

```

IMAGE	ID	DISK USAGE	CONTENT SIZE	In Use	EXTRA
26046660167.dkr.ecr.us-east-1.amazonaws.com/amazon-nginx:v1	ce95c001bce2	584MB	195MB	U	
amazonlinux:latest	ad882807c6c8	227MB	59.3MB		
saved:v1	ce95c001bce2	584MB	195MB	U	
savebms7/techie:v1	ce95c001bce2	584MB	195MB	U	
jenkins/jenkins:lts	01c992ffef29	799MB	289MB	U	
nginx:latest	5aca99593157	241MB	66MB	U	
nginx:v1	9c845c3a54f0	238MB	63.1MB		
sonarqube:latest	35bedac3f40c	2.52GB	1.08GB		
tomcat:latest	12dad7111b0e	589MB	158MB		
ubuntu:latest	f3d28607ddd7	160MB	45.3MB		

```

root@ip-172-31-0-135:/tmp#

```

Now lets run the newlu copies container using command

docker run -itd -p 8080:8080 tomcat:latest

```

root@ip-172-31-0-135:/tmp# docker run -itd -p 8080:8080 tomcat:latest
00cc611a8d74669f1741cb5cd4f54ac2141ad6024638649c4254392b761463de
root@ip-172-31-0-135:/tmp# docker ps

```

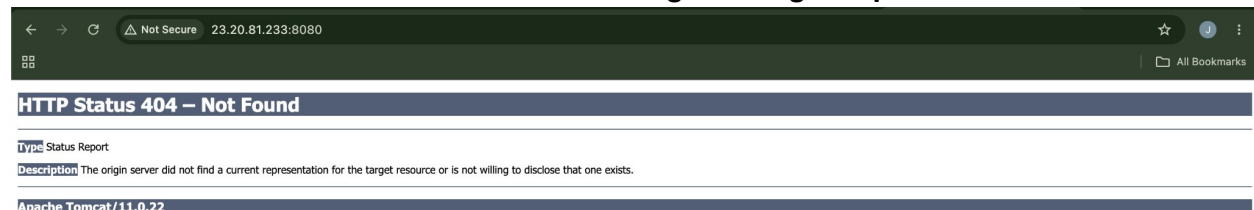
CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS
00cc611a8d74	tomcat:latest	"catalina.sh run"	10 seconds ago	Up 10 s
00cc611a8d74	tomcat:latest	"catalina.sh run"	10 seconds ago	Up 10 s
ccef05956a19	nginx:latest	"/docker-entrypoint...."	53 minutes ago	Up 53 m
ccef05956a19	nginx:latest	"/docker-entrypoint...."	53 minutes ago	Up 53 m

```

root@ip-172-31-0-135:/tmp#

```

Docker Container has been created and working on the given port



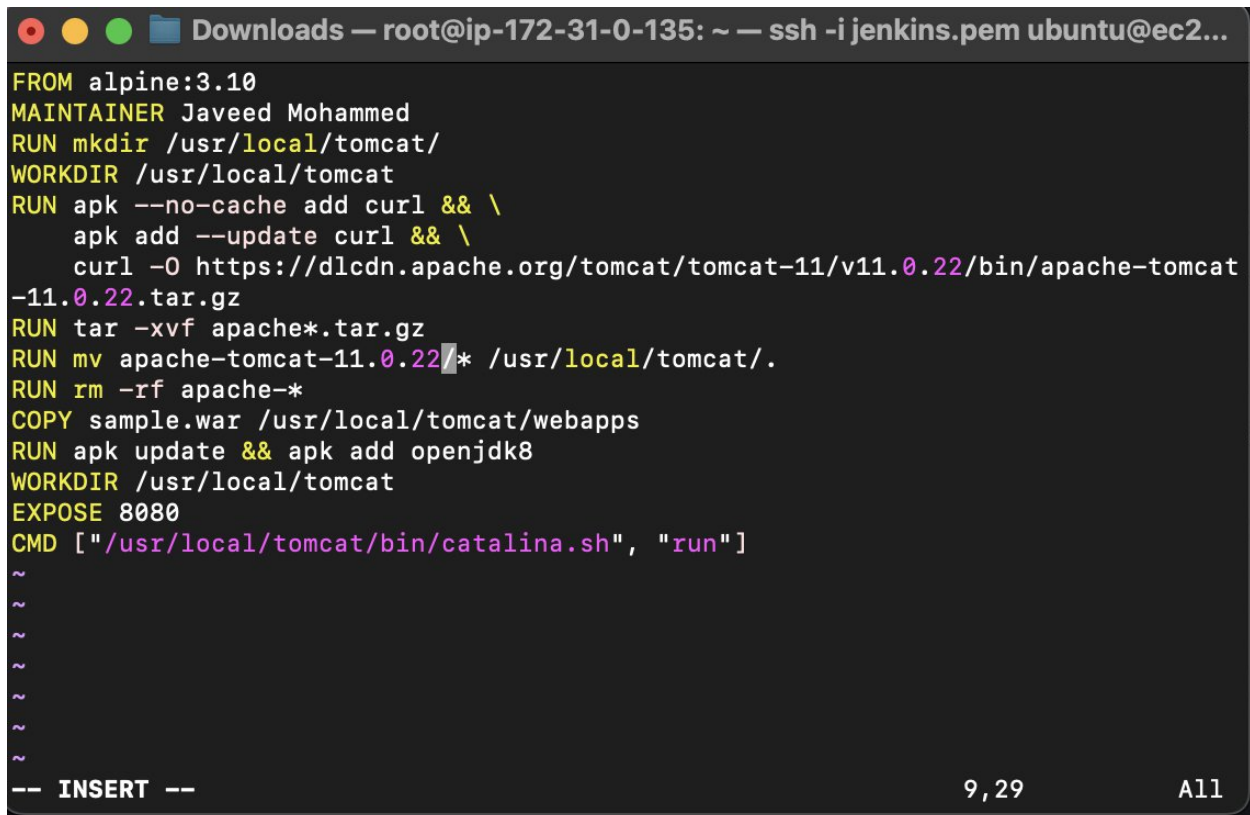
3. Create Docker images using alpine and customize with tomcat.
So lets first create a Dockerfile, Using command

```

vi Dockerfile →FROM alpine:3.10
MAINTAINER Techie_Horizon
RUN mkdir /usr/local/tomcat/
WORKDIR /usr/local/tomcat
RUN apk --no-cache add curl && \
    apk add --update curl && \

```

```
curl -O https://dlcdn.apache.org/tomcat/tomcat-9/v9.0.73/bin/apache-tomcat-9.0.73.tar.gz
RUN tar -xvf apache*.tar.gz
RUN mv apache-tomcat-9.0.73/* /usr/local/tomcat/.
RUN rm -rf apache-*
COPY sample.war /usr/local/tomcat/webapps
RUN apk update && apk add openjdk8
WORKDIR /usr/local/tomcat
EXPOSE 8080
CMD ["/usr/local/tomcat/bin/catalina.sh", "run"]
```



```
FROM alpine:3.10
MAINTAINER Javeed Mohammed
RUN mkdir /usr/local/tomcat/
WORKDIR /usr/local/tomcat
RUN apk --no-cache add curl && \
    apk add --update curl && \
    curl -O https://dlcdn.apache.org/tomcat/tomcat-11/v11.0.22/bin/apache-tomcat-11.0.22.tar.gz
RUN tar -xvf apache*.tar.gz
RUN mv apache-tomcat-11.0.22/* /usr/local/tomcat/.
RUN rm -rf apache-*
COPY sample.war /usr/local/tomcat/webapps
RUN apk update && apk add openjdk8
WORKDIR /usr/local/tomcat
EXPOSE 8080
CMD ["/usr/local/tomcat/bin/catalina.sh", "run"]
~
~
~
~
~
~
~
-- INSERT --
```

Now lets create the image from Docker file
docker build -t tomcat:works .

```
Terminal Shell Edit View Window Help
Downloads — root@ip-172-31-0-135: ~ — ssh -l jenkins.pem ubuntu@ec2-23-20-81-233.compute-1.amazonaws.com — 204x55

root@ip-172-31-0-135:~# docker build -t tomcat:works .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
             Install the buildx component to build images with BuildKit:
             https://docs.docker.com/go/buildx/

Sending build context to Docker daemon 31.23kB
Step 1/13 : FROM alpine:3.10
3.10: Pulling from library/alpine
396c31837116: Pulling fs layer
396c31837116: Download complete
396c31837116: Pull complete
Digest: sha256:451eee8bedcb2f029756dc3e9d73bab8e7943c1ac55cfff3a4861c52a0fdd3e98
Status: Downloaded newer image for alpine:3.10
--> 451eee8bedcb
Step 2/13 : MAINTAINER Javeed Mohammed
--> Running in 1c8504164f2b
--> Removed intermediate container 1c8504164f2b
--> ba059cfc6b54
Step 3/13 : RUN mkdir /usr/local/tomcat/
--> Running in bc017814a934
--> Removed intermediate container bc017814a934
--> 927df5ee8f17
Step 4/13 : WORKDIR /usr/local/tomcat
--> Running in a4e52893a9d9
--> Removed intermediate container a4e52893a9d9
--> 8bae99b48216
Step 5/13 : RUN apk --no-cache add curl && apk add --update curl && curl -O https://dlcdn.apache.org/tomcat/tomcat-11/v11.0.22/bin/apache-tomcat-11.0.22.tar.gz
--> Running in 2581308b7ba3
fetch http://dl-cdn.alpinelinux.org/alpine/v3.10/main/x86_64/APKINDEX.tar.gz
fetch http://dl-cdn.alpinelinux.org/alpine/v3.10/community/x86_64/APKINDEX.tar.gz
(1/4) Installing ca-certificates (20191127-r2)
(2/4) Installing nghttp2-libs (1.39.2-r1)
(3/4) Installing libcurl (7.66.0-r4)
(4/4) Installing curl (7.66.0-r4)
Executing busybox-1.30.1-r5.trigger
Executing ca-certificates-20191127-r2.trigger
OK: 7 MiB in 18 packages
fetch http://dl-cdn.alpinelinux.org/alpine/v3.10/main/x86_64/APKINDEX.tar.gz
fetch http://dl-cdn.alpinelinux.org/alpine/v3.10/community/x86_64/APKINDEX.tar.gz
OK: 7 MiB in 18 packages
% Total % Received % Xferd Average Speed Time Time Time Current
Dload Upload Total Spent Left Speed
100 13.7M 100 13.7M 0 0 22.5M 0 --:--:-- --:--:-- --:--:-- 22.5M
--> Removed intermediate container 2581308b7ba3
--> aacd5b81de1d
Step 6/13 : RUN tar -xvf apache*.tar.gz
--> Running in d301b9892e4b
apache-tomcat-11.0.22/conf/
apache-tomcat-11.0.22/conf/catalina.properties
apache-tomcat-11.0.22/conf/context.xml
apache-tomcat-11.0.22/conf/jaspic-providers.xml
apache-tomcat-11.0.22/conf/jaspic-providers.xsd
apache-tomcat-11.0.22/conf/logging.properties
apache-tomcat-11.0.22/conf/server.xml
apache-tomcat-11.0.22/conf/tomcat-users.xml
```

```
Terminal Shell Edit View Window Help
Downloads — root@ip-172-31-0-135: ~ — ssh -l jenkins.pem ubuntu@ec2-23-20-81-233.compute-1.amazonaws.com — 204x55

Step 6/12 : RUN tar -xvf apache*.tar.gz
--> Using cache
--> 8bae99b48216
Step 7/12 : RUN mv apache-tomcat-11.0.22/* /usr/local/tomcat/.
--> Using cache
--> 177a693485e3
Step 8/12 : RUN rm -rf apache-*
--> Using cache
--> 78b9e84b20f4
Step 9/12 : RUN apk update && apk add openjdk8
--> Running in 14c4deded0b
fetch http://dl-cdn.alpinelinux.org/alpine/v3.10/main/x86_64/APKINDEX.tar.gz
fetch http://dl-cdn.alpinelinux.org/alpine/v3.10/community/x86_64/APKINDEX.tar.gz
v3.10.9-43-g3feb769ea3 [http://dl-cdn.alpinelinux.org/alpine/v3.10/main]
v3.10.6-18-ged79e8de83 [http://dl-cdn.alpinelinux.org/alpine/v3.10/community]
OK: 18344 distinct packages available
(1/46) Installing expat (2.2.8-r0)
(2/46) Installing libbz2 (1.0.6-r7)
(3/46) Installing libpng (1.6.37-r1)
(4/46) Installing freetype (2.10.0-r1)
(5/46) Installing libuuid (2.33.2-r0)
(6/46) Installing fontconfig (2.13.1-r0)
(7/46) Installing encodings (1.0.4-r1)
(8/46) Installing libfontenc (1.1.4-r0)
(9/46) Installing mkfontscale (1.2.1-r1)
(10/46) Installing ttf-dejavu (2.37-r1)
(11/46) Installing libxau (1.0.9-r0)
(12/46) Installing libbsd (0.9.1-r1)
(13/46) Installing libxdmcp (1.1.3-r0)
(14/46) Installing libxcb (1.13.1-r0)
(15/46) Installing libx11 (1.6.12-r1)
(16/46) Installing libxcomposite (0.4.5-r0)
(17/46) Installing libxext (1.3.4-r0)
(18/46) Installing libxi (1.7.9-r2)
(19/46) Installing libxrender (0.9.10-r3)
(20/46) Installing libxtest (1.2.3-r0)
(21/46) Installing atlas-lib (1.1.9-r0)
(22/46) Installing libgcc (8.3.0-r0)
(23/46) Installing giflib (5.1.9-r0)
(24/46) Installing libjpeg-turbo (2.0.4-r1)
(25/46) Installing libstdc++ (8.3.0-r0)
(26/46) Installing openjdk8-jre-lib (8.275.01-r0)
(27/46) Installing java-common (0.2-r0)
(28/46) Installing libffi (3.2.1-r6)
(29/46) Installing p11-kit (0.23.16.1-r1)
(30/46) Installing libtasn1 (4.14-r0)
(31/46) Installing p11-kit-trust (0.23.16.1-r1)
(32/46) Installing java-cacerts (1.0-r0)
(33/46) Installing nss (3.44.3-r0)
(34/46) Installing sqlite-libs (3.28.0-r3)
(35/46) Installing nss (3.44.3-r0)
(36/46) Installing krb5-conf (1.0-r1)
(37/46) Installing libcom_err (1.45.15-r0)
(38/46) Installing keyutils-libs (1.6-r1)
(39/46) Installing libverto (0.3.1-r0)
```

Successfully Build

```
Terminal Shell Edit View Window Help
Downloads — root@ip-172-31-0-135: ~ — ssh -i jenkins.pem ubuntu@ec2-23-20-81-233.compute-1.amazonaws.com — 204x55

(15/46) Installing libx11 (1.6.12-r1)
(16/46) Installing libxcomposite (0.4.5-r0)
(17/46) Installing libxext (1.3.4-r0)
(18/46) Installing libxi (1.7.9-r2)
(19/46) Installing libxrender (0.9.10-r3)
(20/46) Installing libxtst (1.2.3-r3)
(21/46) Installing alsa-lib (1.1.9-r0)
(22/46) Installing libgcc (8.3.0-r0)
(23/46) Installing giflib (5.1.9-r0)
(24/46) Installing libjpeg-turbo (2.0.4-r1)
(25/46) Installing libstdc++ (8.3.0-r0)
(26/46) Installing openjdk8-jre-lib (8.275.01-r0)
(27/46) Installing java-common (0.2-r0)
(28/46) Installing libffi (3.2.1-r6)
(29/46) Installing p11-kit (0.23.16.1-r1)
(30/46) Installing libtasn1 (4.14-r0)
(31/46) Installing p11-kit-trust (0.23.16.1-r1)
(32/46) Installing java-cacerts (1.0-r0)
(33/46) Installing nmap (4.20-r0)
(34/46) Installing sqlite-libs (3.28.0-r3)
(35/46) Installing nss (3.44.3-r0)
(36/46) Installing krb5-conf (1.0-r1)
(37/46) Installing libcom_err (1.46.0-r0)
(38/46) Installing keyutils-libs (1.6-r1)
(39/46) Installing libverto (0.3.1-r0)
(40/46) Installing krb5-libs (1.17.2-r0)
(41/46) Installing lcms2 (2.0-r1)
(42/46) Installing postgres-libs (1.8.25-r1)
(43/46) Installing libk5crypto (1.0.18-r0)
(44/46) Installing openjdk8-jre-base (8.275.01-r0)
(45/46) Installing openjdk8-jre (8.275.01-r0)
(46/46) Installing openjdk8 (8.275.01-r0)
Executing busybox-1.30.1-r5.trigger
Executing ca-certificates-20191127-r2.trigger
Executing fontconfig-2.13.1-r0.trigger
Executing mkfontscale-1.2.1-r1.trigger
Executing java-common-0.2-r0.trigger
OK: 126 MiB in 64 packages
--> Removed intermediate container 14daa9eddd08
--> 4a588fda351
Step 10/12 : WORKDIR /usr/local/tomcat
--> Running in d8ac9ce0e5b1
--> Removed intermediate container d8ac9ce0e5b1
--> 34da60763c1
Step 11/12 : EXPOSE 8080
--> Running in 988692209d6e
--> Removed intermediate container 988692209d6e
--> cd8080b38dc
Step 12/12 : CMD ["./usr/local/tomcat/bin/catalina.sh", "run"]
--> Running in ae940b01b869
--> Removed intermediate container ae940b01b869
--> 9075ca547b96
Successfully built 9075ca547b96
Successfully tagged tomcat:works
root@ip-172-31-0-135:~#
```

By using dockerfile a new image has been created
docker images

```
root@ip-172-31-0-135:~# docker images

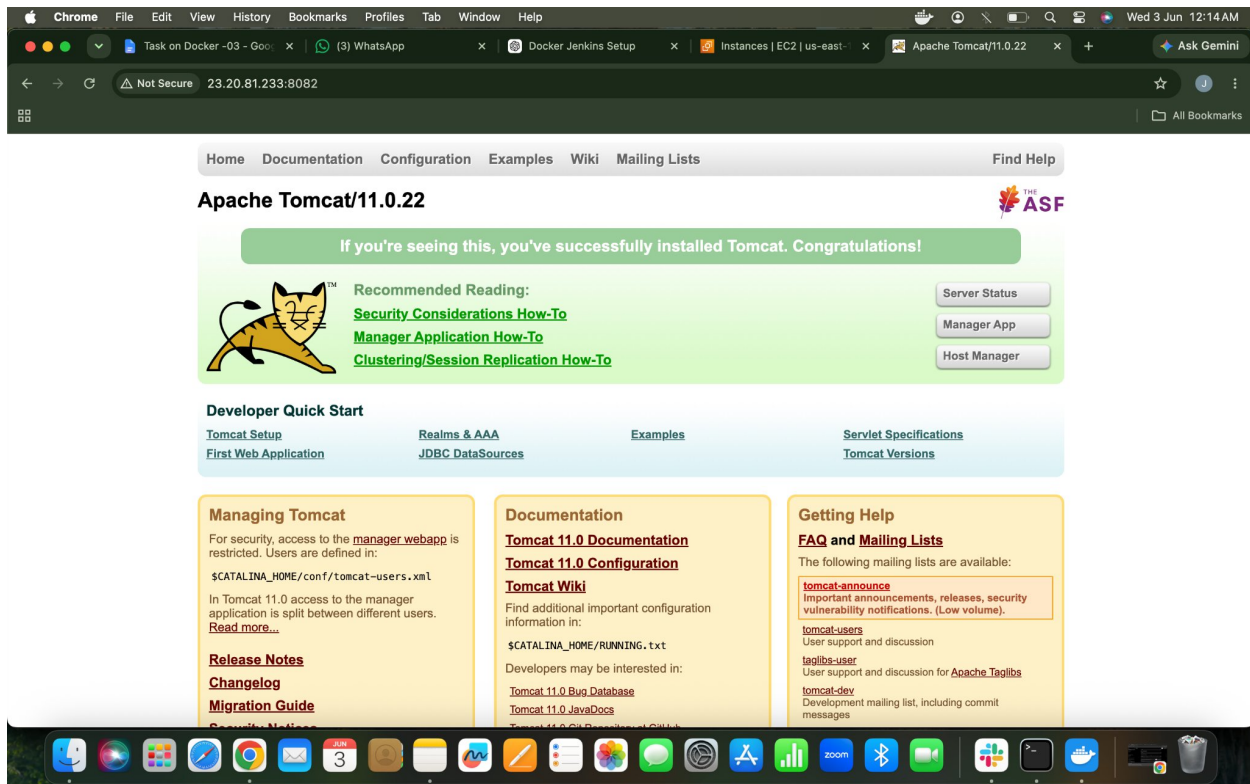
IMAGE                                ID                                DISK USAGE  CONTENT SIZE  EXTRA
92604660167.dkr.ecr.us-east-1.amazonaws.com/amazon-nginx:v1  ce95c001bce2  584MB       195MB        U
alpine:3.10                                                         451eee8bedcb  8.95MB      2.8MB
amazonlinux:latest          ad882807c6c8  227MB       59.3MB
javed:v1                    ce95c001bce2  584MB       195MB        U
javeebms7/techie:v1         ce95c001bce2  584MB       195MB        U
jenkins/jenkins:lts         01c992ffef29  799MB       289MB        U
nginx:latest                5aca99593157  241MB       66MB
nginx:v1                    9c845c3a54f0  238MB       63.1MB
sonarqube:latest            35bedac3f40c  2.52GB      1.08GB
tomcat:latest               12dad7111b0e  589MB       158MB        U
tomcat:works                9075ca547b96  325MB       133MB
ubuntu:latest               f3d28607ddd7  160MB       45.3MB
root@ip-172-31-0-135:~#
```

Lets run and check in the browser using command

docker run -itd -p -8082:8080 tomcat:works

```
root@ip-172-31-0-135:~# docker run -itd -p -8082:8080 tomcat:works
docker: invalid hostPort: -8082

Run 'docker run --help' for more information
[root@ip-172-31-0-135:~# docker run -itd -p 8082:8080 tomcat:works
28185b0718fb17a0845b1e2589145ac3c0181294a7fd54642afe241d34ab4551
root@ip-172-31-0-135:~#
```

The WORKDIR → in dockerfile

```
[root@ip-172-31-0-135:~# docker exec -it 5a5 /bin/sh
/usr/local/tomcat #
```

4. Create single stage and multi stage docker file using this source code:<https://github.com/betawins/multi-stage-example.git>

Single Stage Docker File:

First lets clone or github repo in our machine

```
[root@ip-172-31-0-135:~# git clone https://github.com/betawins/multi-stage-example.git
Cloning into 'multi-stage-example'...
remote: Enumerating objects: 31, done.
remote: Counting objects: 100% (7/7), done.
remote: Compressing objects: 100% (6/6), done.
remote: Total 31 (delta 2), reused 1 (delta 1), pack-reused 24 (from 1)
Receiving objects: 100% (31/31), 53.25 KiB | 6.66 MiB/s, done.
Resolving deltas: 100% (3/3), done.
root@ip-172-31-0-135:~#
```

Once Its cloned do

ls⇒cd multi-stage-example⇒ls⇒vi Dockerfile⇒

FROM maven:3.8.5-openjdk-11

WORKDIR /app

COPY . .

RUN mvn clean package DskipTests

EXPOSE 8080

CMD ["java","-jar","target/*.jar"]

```
[root@ip-172-31-0-135:~# cd multi-stage-example/
[root@ip-172-31-0-135:~/multi-stage-example# cd ..
root@ip-172-31-0-135:~# ls
Docfile  Dockerfile  index.html  multi-stage-example  snap
[root@ip-172-31-0-135:~# cd multi-stage-example/
root@ip-172-31-0-135:~/multi-stage-example# ls
Dockerfile  README.md  mvnw  mvnw.cmd  pom.xml  src
root@ip-172-31-0-135:~/multi-stage-example# vi Dockerfile
```

First lets clone or github repo in our machine


```
[root@ip-172-31-0-135:~/multi-stage-example# docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
alpine:3.19	6baf43584bcb	11.6MB	3.51MB	
maven:3.8.5-openjdk-17	3a9c30b3af62	1.26GB	432MB	
nginx:latest	5aca99593157	241MB	66MB	U
single-stage:v1	a8dccde572b6	1.4GB	493MB	
tomcat:latest	12dad7111b0e	589MB	158MB	U
tomcat:works	d98f3b439aa4	567MB	204MB	U

```
root@ip-172-31-0-135:~/multi-stage-example#
```

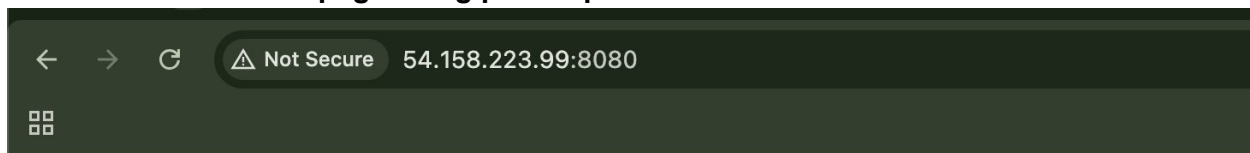
Now lets run the image
docker run -itd -p 8080:8080 --name spring-boot2 single-stage:v2

```
root@ip-172-31-0-135:~/multi-stage-example# docker run -d -p 8080:8080 --name spring-boot2 singlestage:v2
3678f4a8295e2a589358343cfc2ecfa3189cf17d9044728bcb9bd0f2b886ad36
root@ip-172-31-0-135:~/multi-stage-example# docker ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
3678f4a8295e	singlestage:v2	"/usr/local/bin/mvn--"	7 seconds ago	Up 6 seconds	0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp	spring-boot2
33774b9f84b5	singlestage	"/usr/local/bin/mvn--"	6 minutes ago	Exited (1) 6 minutes ago		spring-boot
e808dd0d84f0	31213c9c6cf8	"/bin/sh -c 'mvn cle--"	7 hours ago	Exited (1) 7 hours ago		gallant_pasteur
6aa1111ef87d	single-stage:v1	"/usr/local/bin/mvn--"	7 hours ago	Exited (1) 7 hours ago		elegant_rubin
b8cb793a23a5	single-stage:v1	"/usr/local/bin/mvn--"	7 hours ago	Exited (1) 7 hours ago		adoring_bohr
5e4632a283d7	single-stage:v1	"/usr/local/bin/mvn--"	8 hours ago	Exited (1) 8 hours ago		great_wescoff
915df3f8e7b6	single-stage:v1	"/usr/local/bin/mvn--"	8 hours ago	Created		bold_robinson
994861a032c9	80240f0ba5a5	"/bin/sh -c 'mvn cle--"	8 hours ago	Exited (1) 8 hours ago		brave_antonelli
5a5c7c7d7972	tomcat:works	"/usr/local/tomcat/b--"	8 hours ago	Exited (143) 7 hours ago		frosty_joliot
00cc611a8d74	tomcat:latest	"catalina.sh run"	9 hours ago	Exited (143) 7 hours ago		objective_satoshi
cccf05956a19	nginx:latest	"/docker-entrypoint--"	9 hours ago	Exited (0) 7 hours ago		jolly_elgamal

```
root@ip-172-31-0-135:~/multi-stage-example#
root@ip-172-31-0-135:~/multi-stage-example#
```

Lets check in the web page using public ip:8080



Whitelabel Error Page

This application has no explicit mapping for /error, so you are seeing this as a fallback.

Wed Jun 03 02:46:32 UTC 2026

There was an unexpected error (type=Not Found, status=404).

No message available

Multi-Stage Dockerfile

Lets create a Dockerfile

Stage 1 - Build

FROM maven:3.8.5-openjdk-11 AS build

WORKDIR /app

COPY . .

RUN mvn clean package

Stage 2 - Runtime

FROM eclipse-temurin:11-jre

WORKDIR /app

COPY --from=build /app/target/*.jar app.jar

EXPOSE 8080

CMD ["java", "-jar", "app.jar"]

```
Downloads — root@ip-172-31-0
# Stage 1 - Build
FROM maven:3.8.5-openjdk-11 AS build

WORKDIR /app

COPY . .

RUN mvn clean package -DskipTests

# Stage 2 - Runtime
FROM eclipse-temurin:11-jre

WORKDIR /app

COPY --from=build /app/target/*.jar app.jar

EXPOSE 8080

CMD ["java", "-jar", "app.jar"]
```

Lets Understand whats happing in the above Docker file:

This Dockerfile is using a multi-stage build to package and run a Java application efficiently. In the **first stage (the build stage)**, it uses a **Maven image with OpenJDK 11** to compile the application: it sets a working directory, copies the entire project into the container, and **runs mvn clean package** to generate a **runnable JAR file**. In the **second stage (the runtime stage)**, it uses a much smaller Java Runtime Environment (JRE) image to keep the final

container lightweight. It sets a working directory, **copies only the built JAR file from the first stage into this new image, exposes port 8080** for the application, and finally runs the JAR using `java -jar`. This approach helps **reduce the final image size and improves security by excluding build tools from the runtime environment.**

Lets start the image

docker build -t mutistage:v2 .

```
Terminal Shell Edit View Window Help
Downloads — root@ip-172-31-0-135: ~/multi-stage-example — ssh -i jenkins.pem ubuntu@ec2-54-158-223-99.compute-1.amazonaws.com — 204x54

root@ip-172-31-0-135:~/multi-stage-example# docker build -t mutistage:v1 .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon 211.5Kb
Step 1/9 : FROM maven:3.8.5-openjdk-11 AS build
----> 4e4b4e0a8bc2
Step 2/9 : WORKDIR /app
----> Using cache
----> e0b79c4a4a6c
Step 3/9 : COPY . .
----> 57f3fa1e862b
Step 4/9 : RUN mvn clean package -DskipTests
----> Running in 84e6b7bb24e
[INFO] Scanning for projects...
Downloaded from central: https://repo.maven.apache.org/maven2/org/springframework/boot/spring-boot-starter-parent/2.2.6.RELEASE/spring-boot-starter-parent-2.2.6.RELEASE.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/springframework/boot/spring-boot-starter-parent/2.2.6.RELEASE/spring-boot-starter-parent-2.2.6.RELEASE.pom (8.1 kB at 18 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/springframework/boot/spring-boot-dependencies/2.2.6.RELEASE/spring-boot-dependencies-2.2.6.RELEASE.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/springframework/boot/spring-boot-dependencies/2.2.6.RELEASE/spring-boot-dependencies-2.2.6.RELEASE.pom (127 kB at 1.5 MB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/com/fasterxml/jackson/jackson-bom/2.10.3/jackson-bom-2.10.3.pom
Downloaded from central: https://repo.maven.apache.org/maven2/com/fasterxml/jackson/jackson-bom/2.10.3/jackson-bom-2.10.3.pom (13 kB at 261 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/com/fasterxml/jackson/jackson-parent/2.10/jackson-parent-2.10.pom
Downloaded from central: https://repo.maven.apache.org/maven2/com/fasterxml/jackson/jackson-parent/2.10/jackson-parent-2.10.pom (8.3 kB at 162 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/com/fasterxml/oss-parent/38/oss-parent-38.pom
Downloaded from central: https://repo.maven.apache.org/maven2/com/fasterxml/oss-parent/38/oss-parent-38.pom (23 kB at 512 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/micrometer/micrometer-bom/1.3.6/micrometer-bom-1.3.6.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/micrometer/micrometer-bom/1.3.6/micrometer-bom-1.3.6.pom (6.6 kB at 158 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/io/netty/netty-bom/4.1.48.Final/netty-bom-4.1.48.Final.pom
Downloaded from central: https://repo.maven.apache.org/maven2/io/netty/netty-bom/4.1.48.Final/netty-bom-4.1.48.Final.pom (7.9 kB at 214 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/sonatype/oss/oss-parent/7/oss-parent-7.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/sonatype/oss/oss-parent/7/oss-parent-7.pom (4.8 kB at 127 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/io/projectreactor/reactor-bom/Dysprosium-SR6/reactor-bom-Dysprosium-SR6.pom
Downloaded from central: https://repo.maven.apache.org/maven2/io/projectreactor/reactor-bom/Dysprosium-SR6/reactor-bom-Dysprosium-SR6.pom (4.1 kB at 110 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/io/rsocket/rsocket-bom/1.0.0-RC6/rsocket-bom-1.0.0-RC6.pom
Downloaded from central: https://repo.maven.apache.org/maven2/io/rsocket/rsocket-bom/1.0.0-RC6/rsocket-bom-1.0.0-RC6.pom (3.2 kB at 68 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/logging/log4j/log4j-bom/2.12.1/log4j-bom-2.12.1.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/logging/log4j/log4j-bom/2.12.1/log4j-bom-2.12.1.pom (7.2 kB at 164 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/logging/logging-parent/1/logging-parent-1.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/logging/logging-parent/1/logging-parent-1.pom (3.2 kB at 81 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/apache/18/apache-18.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/apache/18/apache-18.pom (16 kB at 348 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/eclipse/jetty/jetty-bom/9.4.27.v20200227/jetty-bom-9.4.27.v20200227.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/eclipse/jetty/jetty-bom/9.4.27.v20200227/jetty-bom-9.4.27.v20200227.pom (17 kB at 316 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/glassfish/jersey/jersey-bom/2.29.1/jersey-bom-2.29.1.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/glassfish/jersey/jersey-bom/2.29.1/jersey-bom-2.29.1.pom (19 kB at 512 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/eclipse/ee4j/project/1.0.5/project-1.0.5.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/eclipse/ee4j/project/1.0.5/project-1.0.5.pom (13 kB at 338 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/jetbrains/kotlin/kotlin-bom/1.3.71/kotlin-bom-1.3.71.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/jetbrains/kotlin/kotlin-bom/1.3.71/kotlin-bom-1.3.71.pom (8.8 kB at 220 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/jetbrains/kotlinx/kotlinx-coroutines-bom/1.3.5/kotlinx-coroutines-bom-1.3.5.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/jetbrains/kotlinx/kotlinx-coroutines-bom/1.3.5/kotlinx-coroutines-bom-1.3.5.pom (4.8 kB at 134 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/junit/junit-bom/5.5.2/junit-bom-5.5.2.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/junit/junit-bom/5.5.2/junit-bom-5.5.2.pom (4.4 kB at 113 kB/s)
```



```
Terminal Shell Edit View Window Help
Downloads — root@ip-172-31-0-135: ~/multi-stage-example — ssh -i jenkins.pem ubuntu@ec2-54-158-223-99.compute-1.amazonaws.com — 204x54

Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/shared/maven-dependency-tree/3.0.1/maven-dependency-tree-3.0.1.jar (37 kB at 79 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/ow2/asm/asm-util/7.2-beta/asm-util-7.2-beta.jar (81 kB at 174 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/ow2/asm/asm/7.2-beta/asm-7.2-beta.jar (169 kB at 364 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/com/google/guava/guava/19.0/guava-19.0.jar (2.3 MB at 4.3 MB/s)
[INFO] Replacing main artifact with repackaged archive
[INFO] BUILD SUCCESS
[INFO] Total time: 25.332 s
[INFO] Finished at: 2026-06-03T03:17:18Z
[INFO]
--> Removed intermediate container 182851040ff9
--> 6b5a0515c6ae
Step 6/9 : FROM eclipse-temurin:11-jre
11-jre: Pulling from library/eclipse-temurin
945a23abc856: Pulling fs layer
d1f60238b9dc: Pulling fs layer
5eb7a530b067: Pulling fs layer
6f5c5aa4e145: Pulling fs layer
1c24335dd46: Pulling fs layer
11d688399e9: Pulling fs layer
5eb7a530b067: Download complete
1c24335dd46: Download complete
487bdec52ac8: Download complete
11d688399e9: Download complete
863d03f2814e: Download complete
945a23abc856: Download complete
6f5c5aa4e145: Download complete
d1f60238b9dc: Download complete
6f5c5aa4e145: Pull complete
1c24335dd46: Pull complete
945a23abc856: Pull complete
5eb7a530b067: Pull complete
11d688399e9: Pull complete
d1f60238b9dc: Pull complete
Digest: sha256:deed6cef8b63a91f94626429ea31b0366cc34f156b55399b704eeda11a7d2ff
Status: Downloaded newer image for eclipse-temurin:11-jre
--> deed6cef8b63
Step 6/9 : WORKDIR /app
--> Running in b427452433fb
--> Removed intermediate container b427452433fb
--> fc1bdeb28a67
Step 7/9 : COPY --from=build /app/target/*.jar app.jar
--> 87d44e086d7
Step 8/9 : EXPOSE 8080
--> Running in 659428f2243e
--> Removed intermediate container 659428f2243e
--> 2636a33440b5
Step 9/9 : CMD ["java","-jar","app.jar"]
--> Running in d1dd41be55c
--> Removed intermediate container d1dd41be55c
--> 46bca2788b7b
Successfully built 46bca2788b7b
Successfully tagged multistage:v1
```

Build is Successfull

Now lets run the container using command

docker run -d -p 8083:8080 --name spring-boot2 multistage:v2

Docker ps -a to check the running containers

```
root@ip-172-31-0-135:~/multi-stage-example# docker run -d -p 8083:8080 --name spring-boot2 multistage:v2
6d831378a4e7a113d8ebc281835ca921a39197b1d1d8ffa15fa2820d5ac21a
root@ip-172-31-0-135:~/multi-stage-example# docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED       STATUS       PORTS                               NAMES
6d831378a4e   multistage:v1  "/_cacert_entrypoint..." 7 seconds ago Up 7 seconds 0.0.0.0:8083->8080/tcp, [::]:8083->8080/tcp  spring-boot3
9c4791fe342a   f9393c7caaa7   "/bin/sh -c 'mvn cle..." 24 minutes ago Exited (1) 23 minutes ago                                upbeat_satoshi
3678f4ab295e   single-stage:v2 "/usr/local/bin/mvn..." 40 minutes ago Up 40 minutes 0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp  spring-boot2
33774b9684b5   single-stage   "/usr/local/bin/mvn..." 47 minutes ago Exited (1) 47 minutes ago                                spring-boot
e5e8e0d084f8   31213c9c6cf8   "/bin/sh -c 'mvn cle..." 8 hours ago   Exited (1) 8 hours ago                                gallant_pasteur
6aa1111ef87d   single-stage:v1 "/usr/local/bin/mvn..." 8 hours ago   Exited (1) 8 hours ago                                elegant_rubin
d8cb793a23a8   single-stage:v1 "/usr/local/bin/mvn..." 8 hours ago   Exited (1) 8 hours ago                                adoring_bohr
5ea432a283d7   single-stage:v1 "/usr/local/bin/mvn..." 8 hours ago   Exited (1) 8 hours ago                                great_wescott
915df33f0e7b   single-stage:v1 "/usr/local/bin/mvn..." 8 hours ago   Created                                             bold_robinson
994861a832c9   8b24bf0ba5a5   "/bin/sh -c 'mvn cle..." 8 hours ago   Exited (1) 8 hours ago                                brave_antonelli
5a5c7c7d7972   tomcat:works   "/usr/local/tomcat/b..." 9 hours ago   Exited (143) 8 hours ago                                frosty_joliot
00cc011a8f74   tomcat:latest   "catalina.sh run"        9 hours ago   Exited (143) 8 hours ago                                objective_satoshi
cce9f0596a19   nginx:latest    "/docker-entrypoint..." 10 hours ago   Exited (0) 8 hours ago                                jolly_elgamal
root@ip-172-31-0-135:~/multi-stage-example#
```

So container is successful created and running

Now lets understand the main difference between single stage and multi stage docker file from the below image

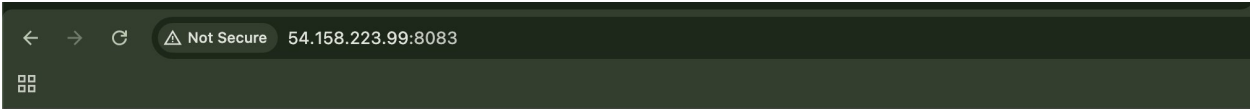
```
3678f4a8295e  spring-boot2  0.15%  182.4MiB / 1.86GiB  9.57%  13.2kB / 11.5kB  32.8kB / 1.05MB  35

[^Croot@ip-172-31-0-135:~/multi-stage-example# docker images

IMAGE                ID                DISK USAGE  CONTENT SIZE  EXTRA
alpine:3.19          6baf43584bcb     11.6MB      3.51MB
eclipse-temurin:11-jre  deed6cefb063     430MB       114MB
maven:3.8.5-openjdk-11  4e4b4e0a8bc2     1.05GB      344MB
maven:3.8.5-openjdk-17  3a9c30b3af62     1.26GB      432MB
mutistage:v1          46bca2788b7b     461MB       126MB      U
nginx:latest          5aca99593157     241MB       66MB       U
single-stage:v1       a8dccde572b6     1.4GB       493MB      U
singlestage:latest    576bec4ae05e     1.4GB       493MB      U
singlestageiv2        faa9ca8f6add     1.2GB       405MB      U
tomcat:latest         12dad7111b0e     589MB       158MB      U
tomcat:works          d98f3b439aa4     567MB       204MB      U
root@ip-172-31-0-135:~/multi-stage-example#
```

We can see how much size difference there is when we use multistage.

Let's check in browser too



Whitelabel Error Page

This application has no explicit mapping for /error, so you are seeing this as a fallback.

Wed Jun 03 03:28:23 UTC 2026

There was an unexpected error (type=Not Found, status=404).

No message available

5. Install docker compose and execute sample application.

Lets update the packages first

apt update

```
root@ip-172-31-0-135:~# apt update
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease [136 kB]
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease
Get:4 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64v3 Packages [197 kB]
Get:5 http://security.ubuntu.com/ubuntu resolute-security InRelease [136 kB]
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [51.5 kB]
Get:7 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64 c-n-f Metadata [2992 B]
Get:8 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/universe amd64v3 Packages [98.8 kB]
Get:9 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/universe Translation-en [29.6 kB]
Get:10 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 c-n-f Metadata [392 B]
Get:11 http://security.ubuntu.com/ubuntu resolute-security/main amd64v3 Packages [182 kB]
Get:12 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [49.0 kB]
Get:13 http://security.ubuntu.com/ubuntu resolute-security/universe amd64v3 Packages [94.2 kB]
Get:14 http://security.ubuntu.com/ubuntu resolute-security/universe Translation-en [28.3 kB]
Get:15 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64v3 Packages [193 kB]
Get:16 http://security.ubuntu.com/ubuntu resolute-security/restricted Translation-en [34.2 kB]
Fetched 1232 kB in 1s (2268 kB/s)
11 packages can be upgraded. Run 'apt list --upgradable' to see them.
root@ip-172-31-0-135:~#
```

Lets install docker compose now

```
sudo curl -SL https://github.com/docker/compose/releases/latest/download/docker-
compose-linux-x86_64 \
-o /usr/local/bin/docker-compose
```

```
root@ip-172-31-0-135:~# curl -L https://github.com/docker/compose/releases/download/1.21.0/docker-compose -s -o /usr/local/bin/docker-compose > /dev/null
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           % Done    0     0     0     0     0     0     0
100 10.35M 100 10.35M    0     0  55.46M    0     0     0
```

Now lets check the version of our docker-compose

```
root@ip-172-31-0-135:~# docker-compose --version
-bash: /usr/local/bin/docker-compose: Permission denied
root@ip-172-31-0-135:~#
```

As its says we dont have permission lets give the permissions and check

sudo chmod +x /usr/local/bin/docker-compose and

also lets create sym link using the command

ln -s /usr/local/bin/docker-compose /usr/bin/docker-compose

And now lets check for version

Docker-compose -version

```
root@ip-172-31-0-135:~# docker-compose --version
-bash: /usr/local/bin/docker-compose: Permission denied
root@ip-172-31-0-135:~# sudo chmod +x /usr/local/bin/docker-compose
root@ip-172-31-0-135:~# ln -s /usr/local/bin/docker-compose /usr/bin/docker-compose
root@ip-172-31-0-135:~# docker-compose --version
docker-compose version 1.21.0, build 5920eb0
root@ip-172-31-0-135:~#
```

6 Implement a solution to scan images

Lets execute a sample application now

Inside vi docker-compose.yml;

version: '3'

services:

web:

image: nginx

ports:

- "8080:80"

db:

image: mysql

environment:

MYSQL_ROOT_PASSWORD: root123

```

version: '3'

services:
  web:
    image: nginx
    ports:
      - "8080:80"

  db:
    image: mysql
    environment:
      MYSQL_ROOT_PASSWORD: root123

```

Now lets execute the compose file

docker-compose up -d

```

root@ip-172-31-0-135:~# docker-compose up -d
WARN[0000] /root/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion
[*] up 26/26
✓ Image mysql Pulled 15.8s
✓ Image nginx Pulled 0.3s
✓ Network root_default Created 0.1s
✓ Container root-web-1 Started 0.7s
✓ Container root-db-1 Started 0.7s
root@ip-172-31-0-135:~#

```

docker ps ⇒ for checking the running containers

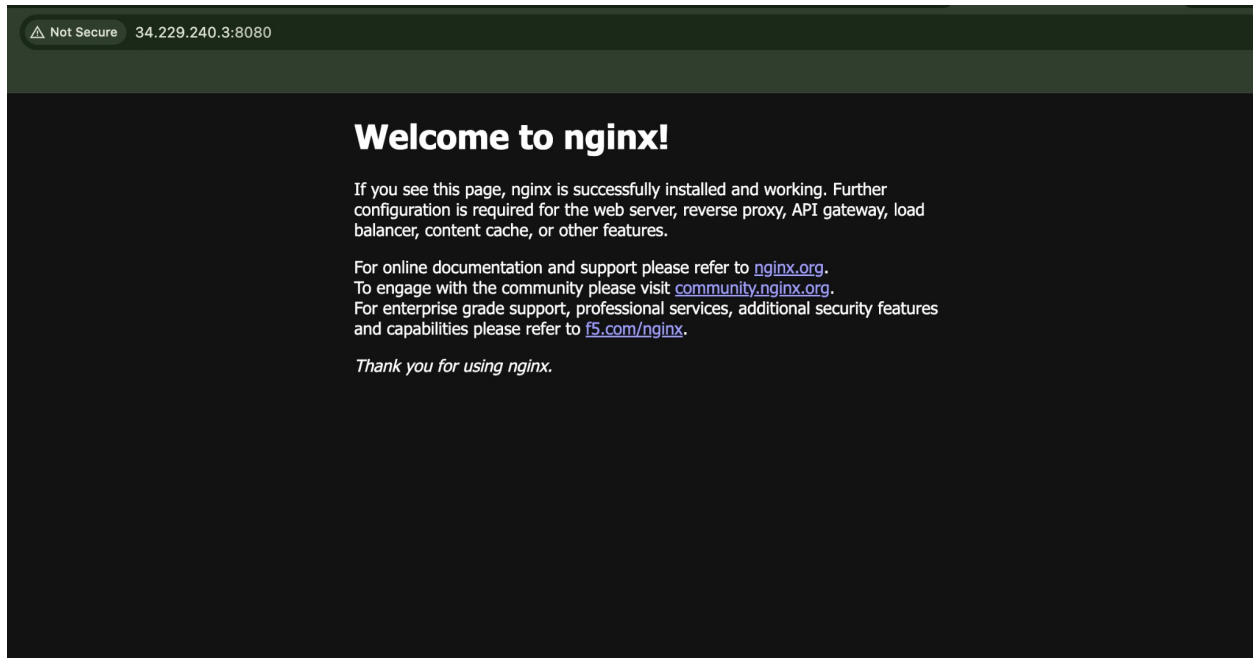
```

root@ip-172-31-0-135:~# docker compose ps
docker: unknown command: docker compose

Run 'docker --help' for more information
root@ip-172-31-0-135:~# docker-compose ps
WARN[0000] /root/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion
NAME      IMAGE      COMMAND                  SERVICE    CREATED   STATUS    PORTS
root-db-1 mysql      "docker-entrypoint.s..." db         3 minutes ago Up 3 minutes 3306/tcp, 33060/tcp
root-web-1 nginx      "/docker-entrypoint..." web        3 minutes ago Up 3 minutes 0.0.0.0:8080->80/tcp, [::]:8080->80/tcp
root@ip-172-31-0-135:~#

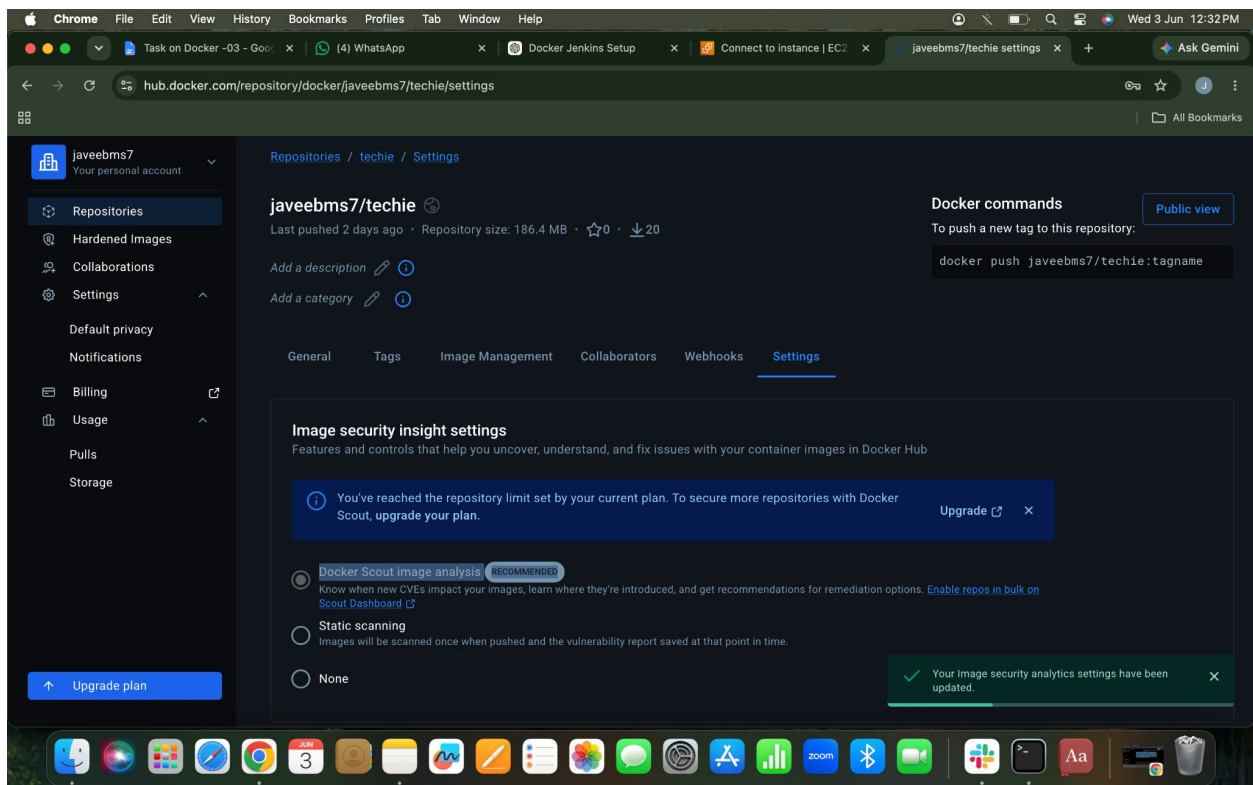
```

Lets check in browser using publicIP:8080



6. Implement a solution to scan images when pushed to the docker registry.
First lets turn on the security option in our docker hub

Docker hub→repository→our repo→settings→**Docker Scout image analysis**→save



Now lets build an image from scratch and push the image to the hub and scan it
So I already have an image i will use the image and push the image to docker hub

```
root@web-1 nginx "/docker-entrypoint..." web 3 minutes ago Up 3 minutes 0.0.0.0:8080->80/t
root@ip-172-31-0-135:~# docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
mutistage:v1	46bca2788b7b	461MB	126MB	U
mysql:latest	c11782aa2a96	1.3GB	290MB	U
nginx:latest	5aca99593157	241MB	66MB	U

```
root@ip-172-31-0-135:~#
```

Docker login ⇒ once succeeded

docker tag mutistage:v1 javeebms/techie:v1

Docker images

Docker push javeebms/techie:v1

```
root@ip-172-31-0-135:~# docker login
Authenticating with existing credentials... [Username: javeebms7]

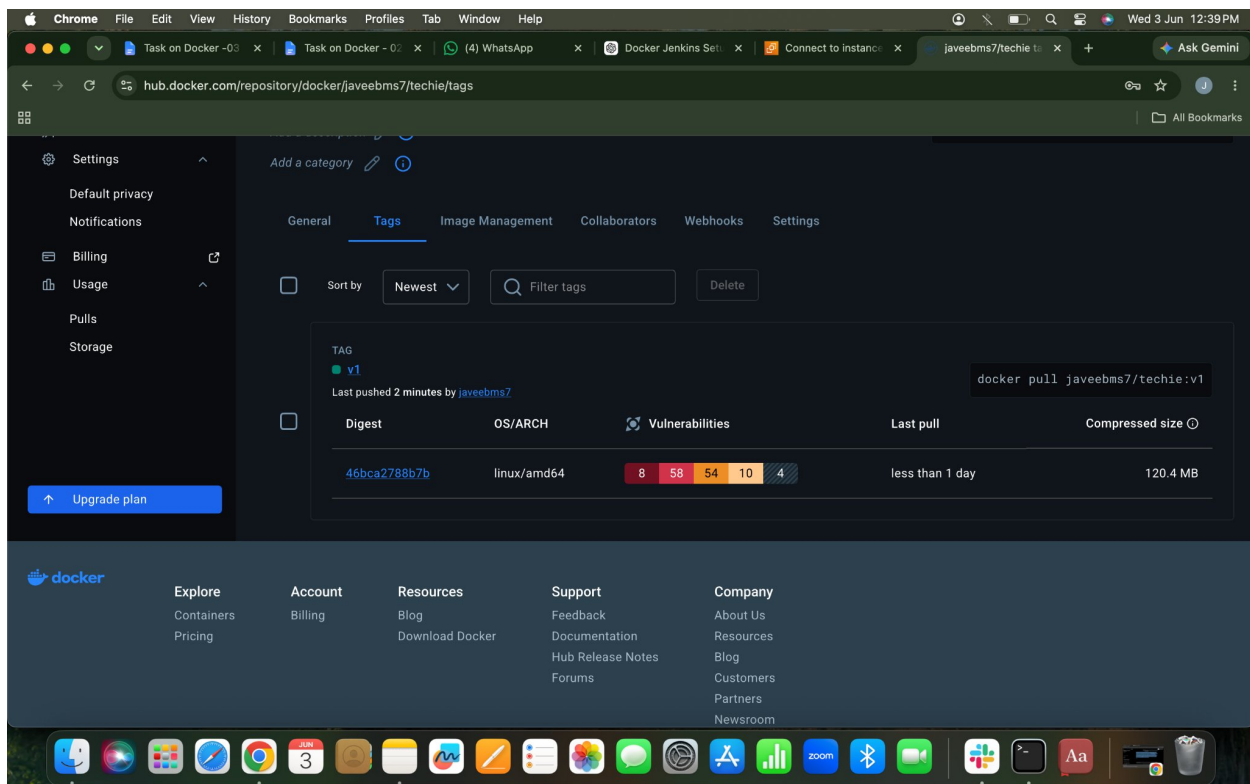
Info → To login with a different account, run 'docker logout' followed by 'docker login'

Login Succeeded
root@ip-172-31-0-135:~# docker tag mutistage:v1 javeebms7/techie:v1
root@ip-172-31-0-135:~# docker images
```

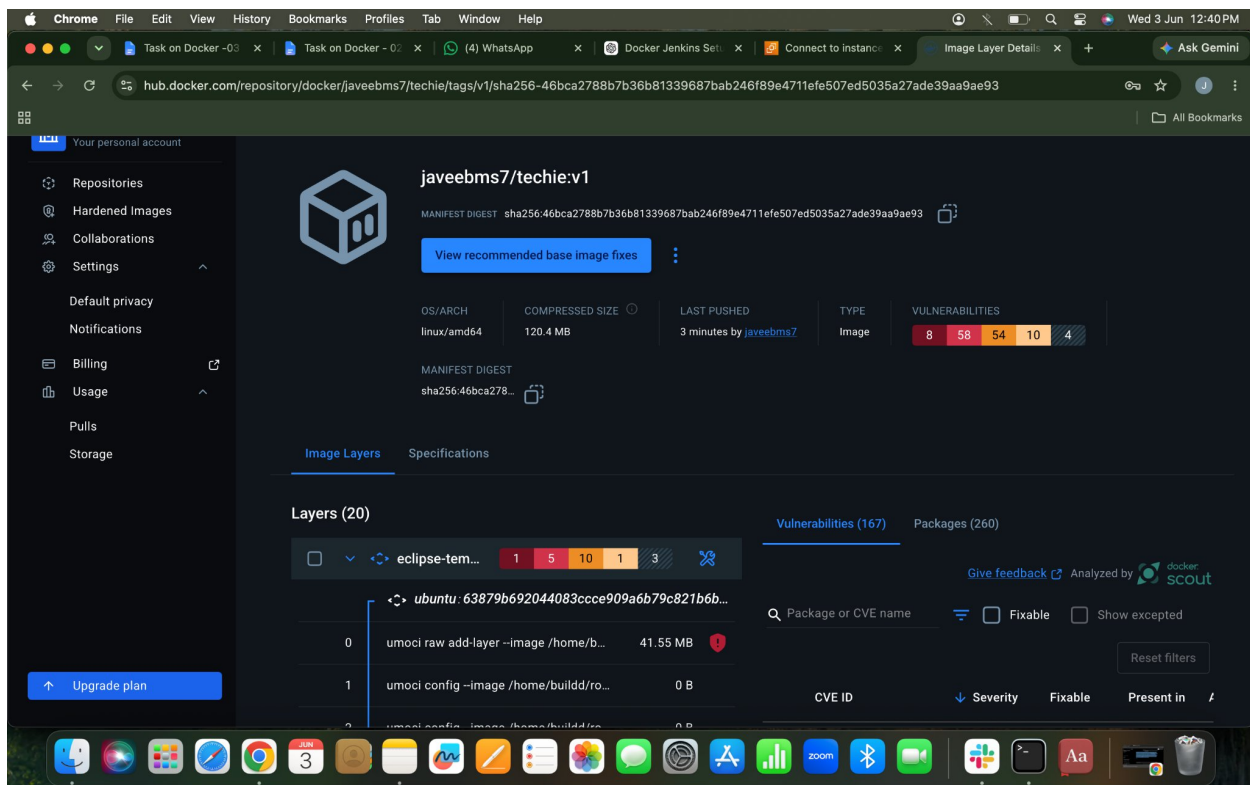
IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
javeebms7/techie:v1	46bca2788b7b	461MB	126MB	U
mutistage:v1	46bca2788b7b	461MB	126MB	U
mysql:latest	c11782aa2a96	1.3GB	290MB	U
nginx:latest	5aca99593157	241MB	66MB	U

```
root@ip-172-31-0-135:~# docker push javeebms7/techie:v1
The push refers to repository [docker.io/javeebms7/techie]
11d6883999e9: Mounted from library/eclipse-temurin
b372c4b29dff: Pushed
b03acb02fc5a: Pushed
6f5c5aa4e145: Mounted from library/eclipse-temurin
1c24335ddd46: Mounted from library/eclipse-temurin
945a23ebc056: Mounted from library/eclipse-temurin
5eb7a5306067: Mounted from library/eclipse-temurin
d1f60238b9dc: Mounted from library/eclipse-temurin
v1: digest: sha256:46bca2788b7b36b81339687bab246f89e4711efe507ed5035a27ade39aa9ae93 size: 2051
root@ip-172-31-0-135:~#
```

Lets check in the Docker Hub



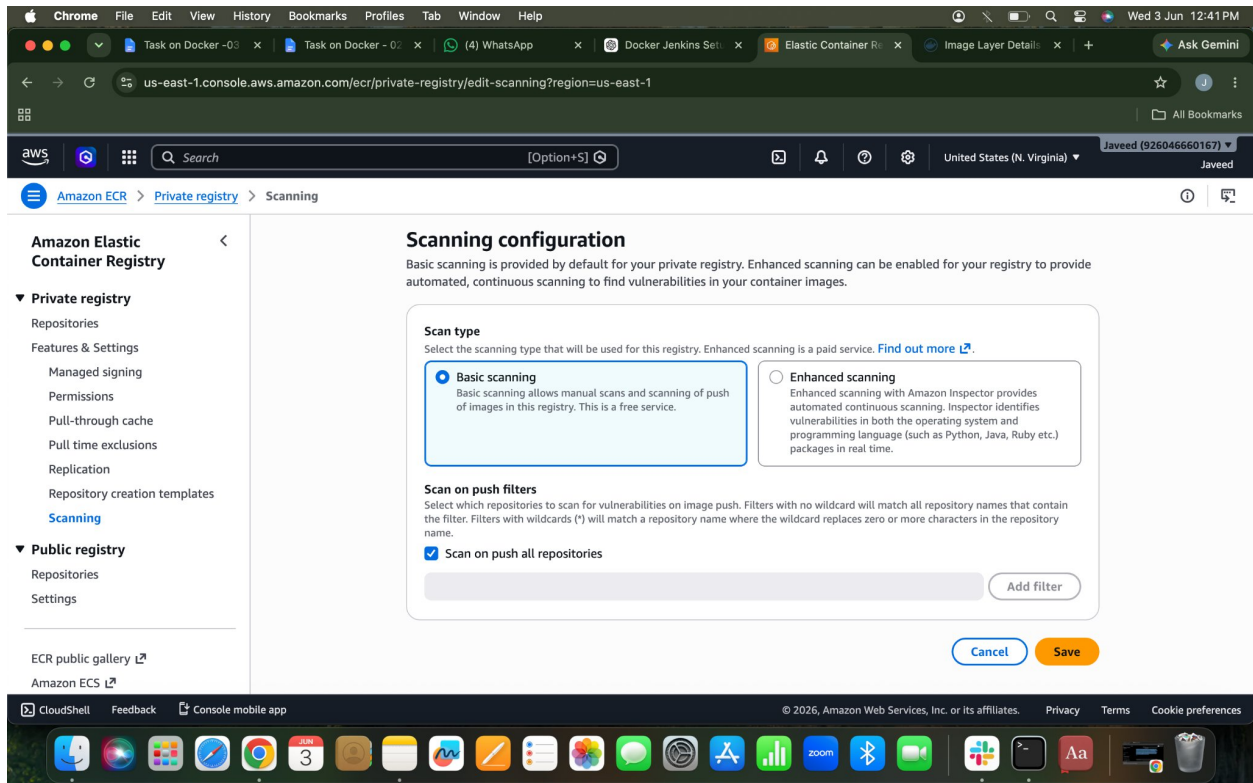
So in Vulnerabilities section we can see the scanned



7. Implement a solution to scan images when pushed to AWS ECR.

Lets turn on the scan feature first

AWS ECR→repo→actions→repo scan filters→basic-save



Now lets goto cli and login to ECR-AWS

`aws ecr get-login-password --region us-east-1 | \`

`docker login --username AWS \`

`--password-stdin 926046660167.dkr.ecr.us-east-1.amazonaws.com`

```
root@ip-172-31-0-135:~# aws ecr get-login-password --region us-east-1 | \
docker login --username AWS \
[--password-stdin 926046660167.dkr.ecr.us-east-1.amazonaws.com

WARNING! Your credentials are stored unencrypted in '/root/.docker/config.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

Login Succeeded
root@ip-172-31-0-135:~#
```

AWS ECR login ⇒ once succeeded

`docker tag mutistage:v1 926046660167.dkr.ecr.us-east-1.amazonaws.com/amazon-nginx:v1`

Docker images

Docker push 926046660167.dkr.ecr.us-east-1.amazonaws.com/amazon-nginx:v1

```

Default output format [json]:
root@ip-172-31-0-135:~# docker images

IMAGE                                ID                                DISK USAGE  CONTENT SIZE  EXTRA
javeebms7/techie:v1                 46bca2788b7b                     461MB       126MB         U
mutistage:v1                         46bca2788b7b                     461MB       126MB         U
mysql:latest                         c11782aa2a96                     1.3GB       290MB         U
nginx:latest                         5aca99593157                     241MB       66MB          U
root@ip-172-31-0-135:~# docker tag m
mutistage:v1 mysql:latest
root@ip-172-31-0-135:~# docker tag mutistage:v1 926046660167.dkr.ecr.us-east-1.amazonaws.com/amazon-nginx:v1
root@ip-172-31-0-135:~# docker images

IMAGE                                ID                                DISK USAGE  CONTENT SIZE  EXTRA
926046660167.dkr.ecr.us-east-1.amazonaws.com/amazon-nginx:v1 46bca2788b7b                     461MB       126MB         U
javeebms7/techie:v1                 46bca2788b7b                     461MB       126MB         U
mutistage:v1                         46bca2788b7b                     461MB       126MB         U
mysql:latest                         c11782aa2a96                     1.3GB       290MB         U
nginx:latest                         5aca99593157                     241MB       66MB          U
root@ip-172-31-0-135:~# docker push 926046660167.dkr.ecr.us-east-1.amazonaws.com/amazon-nginx:v1
The push refers to repository [926046660167.dkr.ecr.us-east-1.amazonaws.com/amazon-nginx]
5eb7a5306067: Waiting
11d6883999e9: Waiting
b372c4b29dff: Waiting
b03acb02fc5a: Waiting
6f5c5aa4e145: Waiting
1c24335ddd46: Waiting
945a23ebc056: Waiting
d1f60238b9dc: Waiting
unknown: unexpected status from HEAD request to https://926046660167.dkr.ecr.us-east-1.amazonaws.com/v2/amazon-nginx/blobs/sha256:5eb7a53060679f1
14f40d1c4a21bcd0e3b5896b71a0: 403 Forbidden
root@ip-172-31-0-135:~# aws sts get-caller-identity
{
  "UserId": "926046660167",
  "Account": "926046660167",
  "Arn": "arn:aws:iam::926046660167:root"
}
root@ip-172-31-0-135:~# aws ecr get-login-password --region us-east-1 | \
docker login --username AWS \
--password-stdin 926046660167.dkr.ecr.us-east-1.amazonaws.com

WARNING! Your credentials are stored unencrypted in '/root/.docker/config.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

Login Succeeded
root@ip-172-31-0-135:~# docker push 926046660167.dkr.ecr.us-east-1.amazonaws.com/amazon-nginx:v1
The push refers to repository [926046660167.dkr.ecr.us-east-1.amazonaws.com/amazon-nginx]
b03acb02fc5a: Pushed
945a23ebc056: Pushed
d1f60238b9dc: Pushed
6f5c5aa4e145: Pushed
5eb7a5306067: Pushed

```

Now lets check in AWS-ECR

Amazon Elastic Container Registry

amazon-nginx

Summary Images Lifecycle policy Permissions Repository tags

Images (2) Info

Filter active images

☐	Image tags	Type	Created at	Image size (MB)	Image digest	Last pulled at
☐	v1	Image	03 June 2026, 12:47:32 (UTC+05.5)	126.25	sha256:46bca...	-
☐	-	Image	01 June 2026, 18:28:48 (UTC+05.5)	195.48	sha256:ce95c...	03 June 2026, 12:47:32 (UTC+05.5)

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Now lets check if the scanning has been done or not
Its in process

Amazon Elastic Container Registry

amazon-nginx

sha256:46bca2788b7b36b81339687bab246f89e4711efe507ed5035a27ade39aa9ae93

General information

Artifact type Image	Repository amazon-nginx	Pushed at 03 June 2026, 12:47:32 (UTC+05.5)
Last recorded pull time -	Size (MB) 126.25	Image status Active

Scanning and vulnerabilities

Scanning image

Status
In progress

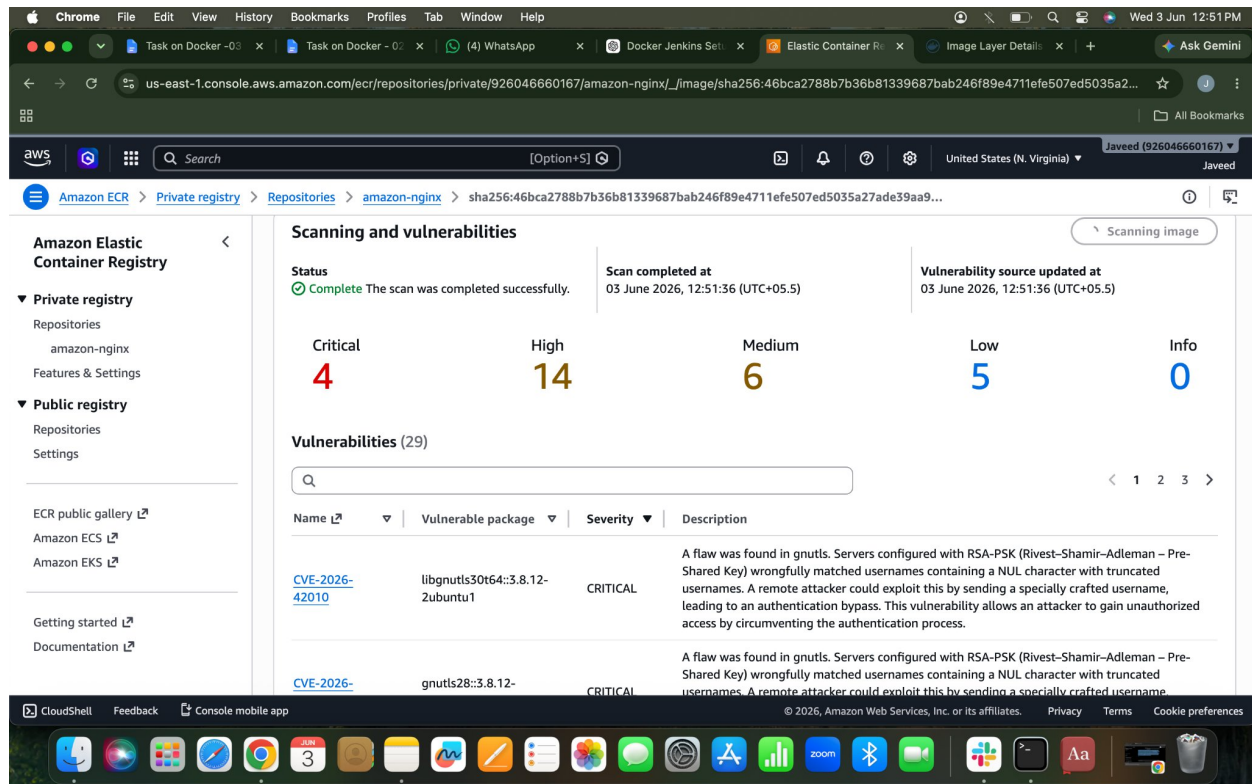
Polling for scan results
The page will automatically update with scan results. This may take several minutes.

Referrers Info

CloudShell Feedback Console mobile app

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Scan Complete



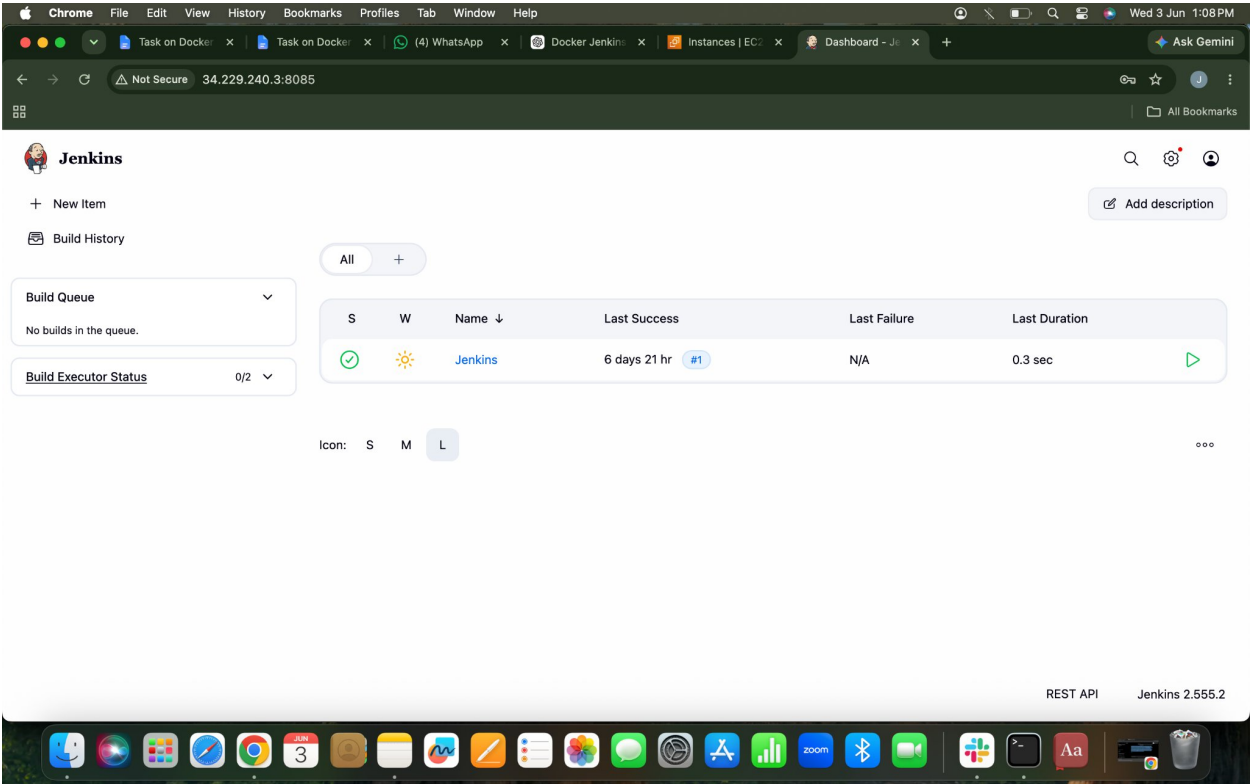
8. Create a Jenkins pipeline to create a docker image and push the image to Docker hub.
So First Lets pull jenkins

```
docker run -itd \  
--name jenkins \  
-p 8085:8080 \  
-p 50000:50000 \  
Jenkins:2.60.3
```

Lets Start jenkins
docker start jenkins

```
root@ip-172-31-0-135:~# docker run -itd \
--name jenkins \
-p 8085:8080 \
-p 50000:50000 \
-v jenkins_home:/var/jenkins_home \
jenkins/jenkins:lts
Unable to find image 'jenkins/jenkins:lts' locally
lts: Pulling from jenkins/jenkins
3bdbd616c406: Pull complete
307f8152a55e: Pull complete
cebc1e77ceba: Pull complete
6b35bb76b8c8: Pull complete
801814f62c99: Pull complete
c2314d42cca1: Pull complete
76a355a69c2e: Pull complete
5834ae4b8774: Pull complete
ae218d1cfa6e: Pull complete
9014a3401c36: Pull complete
dd00b4ea1f64: Pull complete
4c140080734e: Pull complete
653a4e57bccf: Download complete
Digest: sha256:01c992ffef29dcf41c7164e8c16285c657c2368c4943dda8d68c93fdf54447d5
Status: Downloaded newer image for jenkins/jenkins:lts
fc9dd9fede8fc0e30ecfcbbba685bc3ae1c39d1b99f89bc2b7a6a5b9c03c5e694
root@ip-172-31-0-135:~#
```

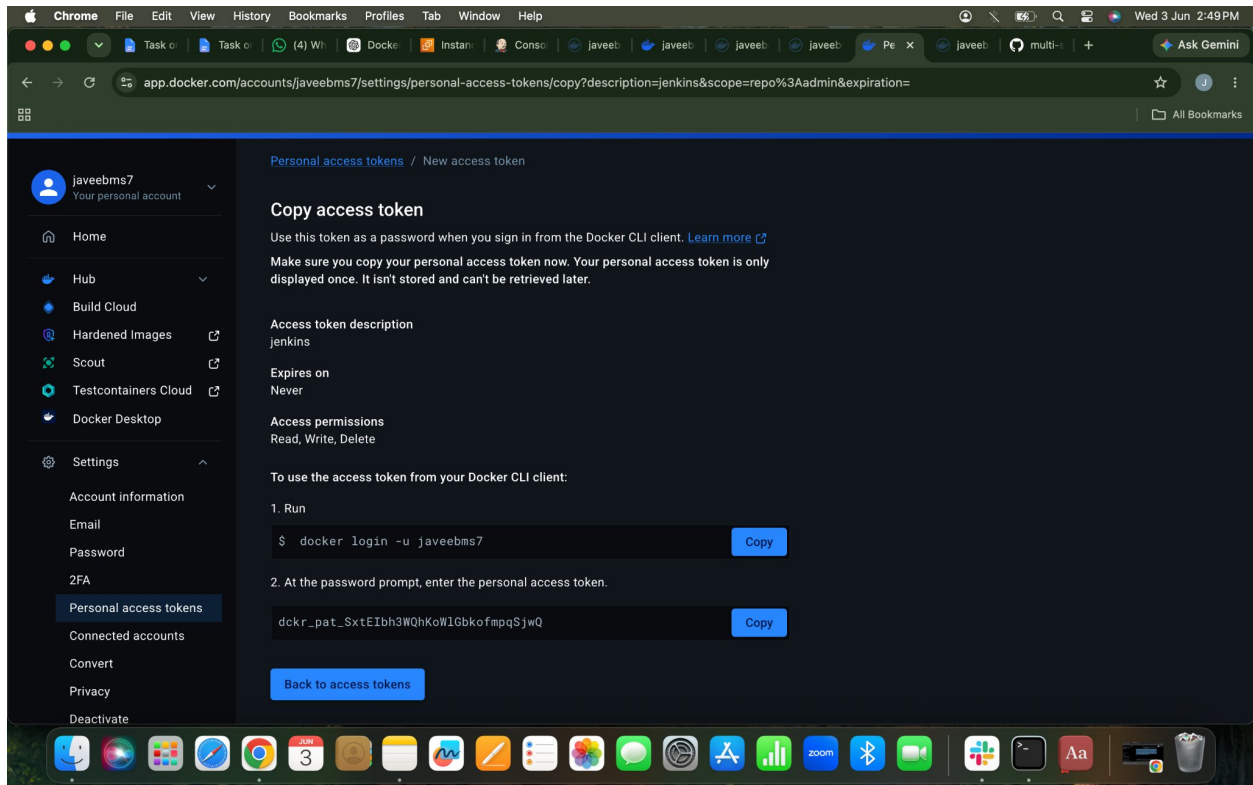
Check publicIP:8085 →adminpasswd→add new user→plugins→



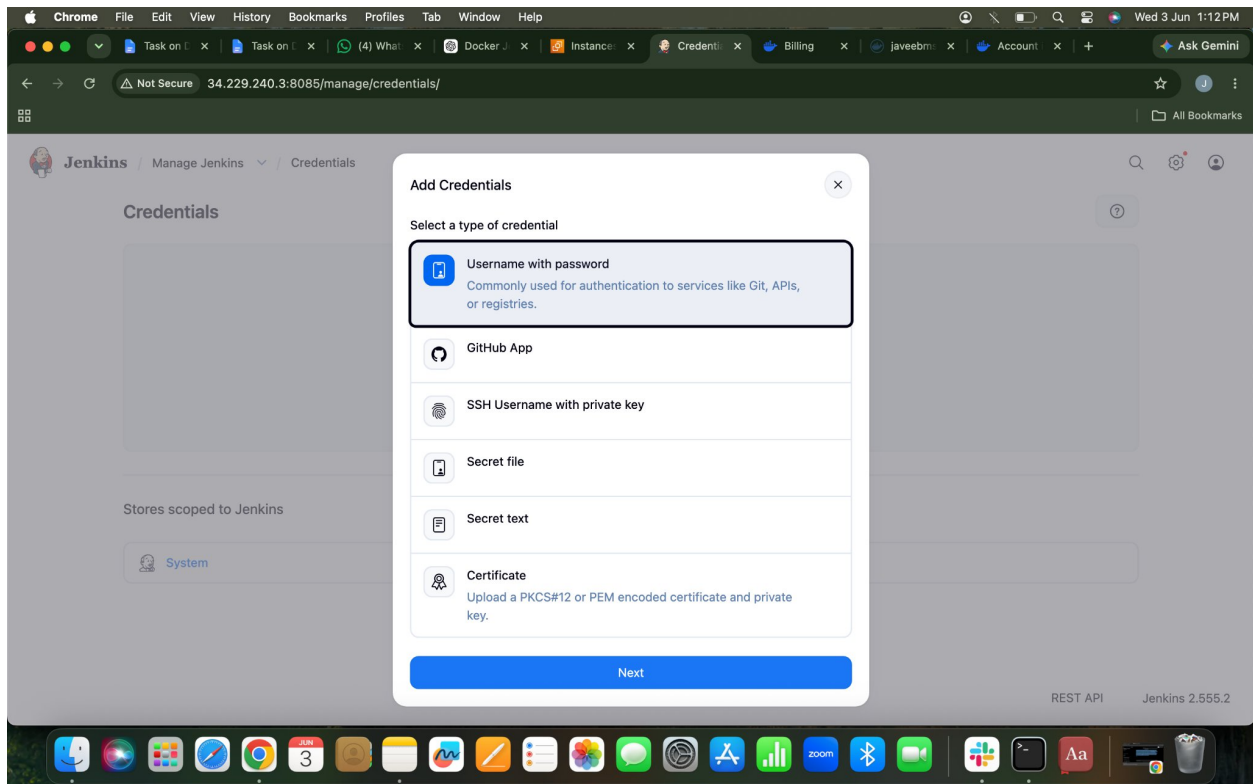
Lets create credentials for Dockerhub

Generate access token in Dockerhub

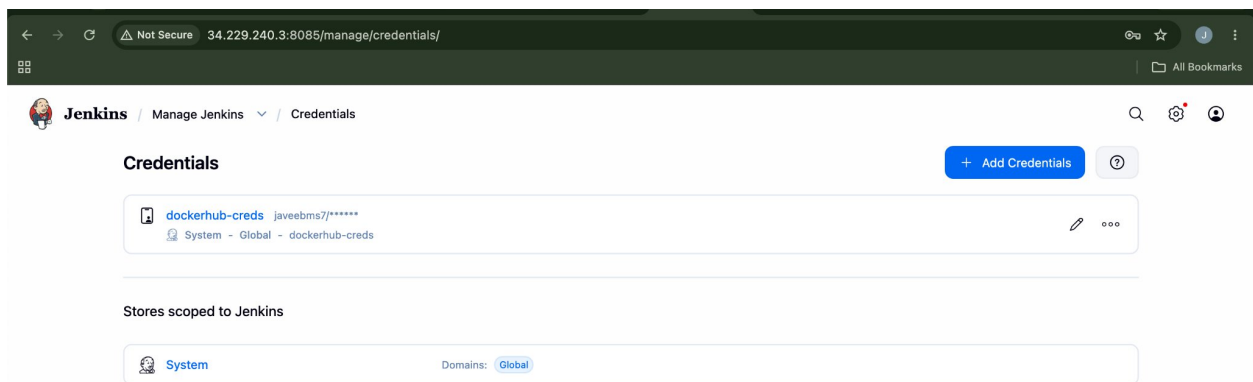
DockerHub→Settings→personal access tokens→permissions(RWX) →copy 2→give in
manage jenkins



Manage jenkins→credentials→



USE Dockerhub Uname and Passwd LEts



**Lets fork the repo and change the Dockerfile, because its using the old versions of java
FROM maven:3.8.6-openjdk-8 AS builder**

RUN mkdir -p /app/source

COPY . /app/source

WORKDIR /app/source

RUN mvn clean package

Stage 2 - Runtime

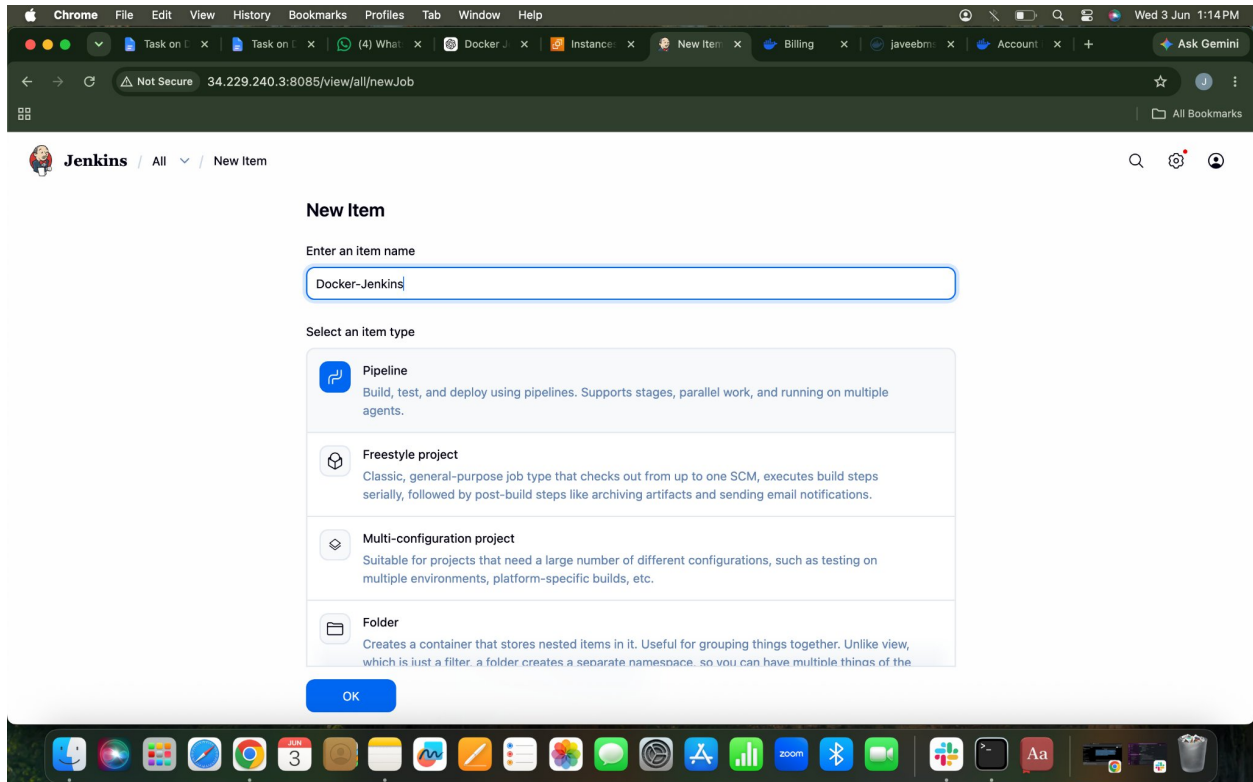
FROM eclipse-temurin:8-jre

COPY --from=builder /app/source/target/*.jar /app/app.jar

EXPOSE 8080

ENTRYPOINT ["java","-Djava.security.egd=file:/dev/./urandom","-jar","/app/app.jar"]

Lets Create new Job



Add Script

pipeline {

agent any

environment {

IMAGE_NAME = "javeebms7/techie"

IMAGE_TAG = "v1"

}

stages {

stage('Clone Code') {

steps {

git 'https://github.com/javeebms7/multi-stage-example.git'

}

}

stage('Build Docker Image') {

steps {

sh 'docker build -t \$IMAGE_NAME:\$IMAGE_TAG .'

}

}

```

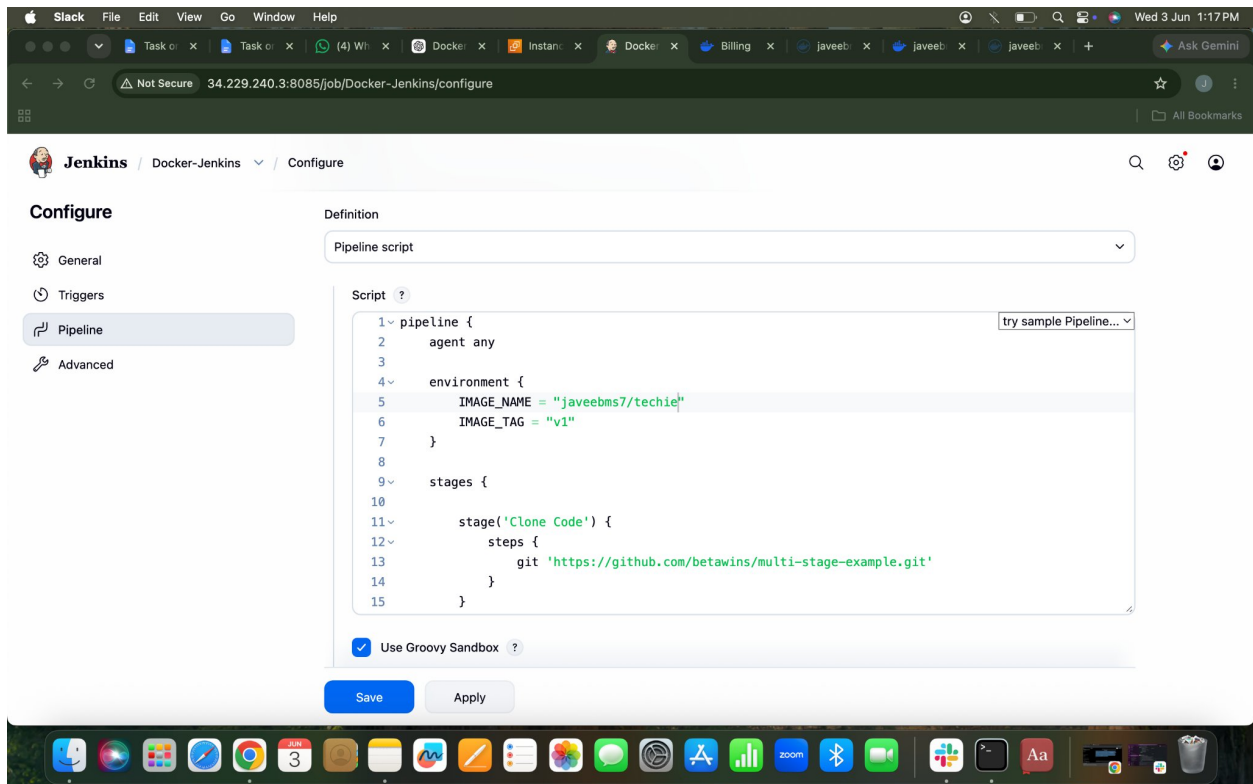
stage('Login DockerHub') {
    steps {
        withCredentials([
            usernamePassword(
                credentialsId: 'dockerhub-creds',
                usernameVariable: 'DOCKER_USER',
                passwordVariable: 'DOCKER_PASS'
            )
        ]) {
            sh """
            echo $DOCKER_PASS | docker login \
            -u $DOCKER_USER \
            --password-stdin
            """
        }
    }
}

stage('Push Image') {
    steps {
        sh 'docker push $IMAGE_NAME:$IMAGE_TAG'
    }
}

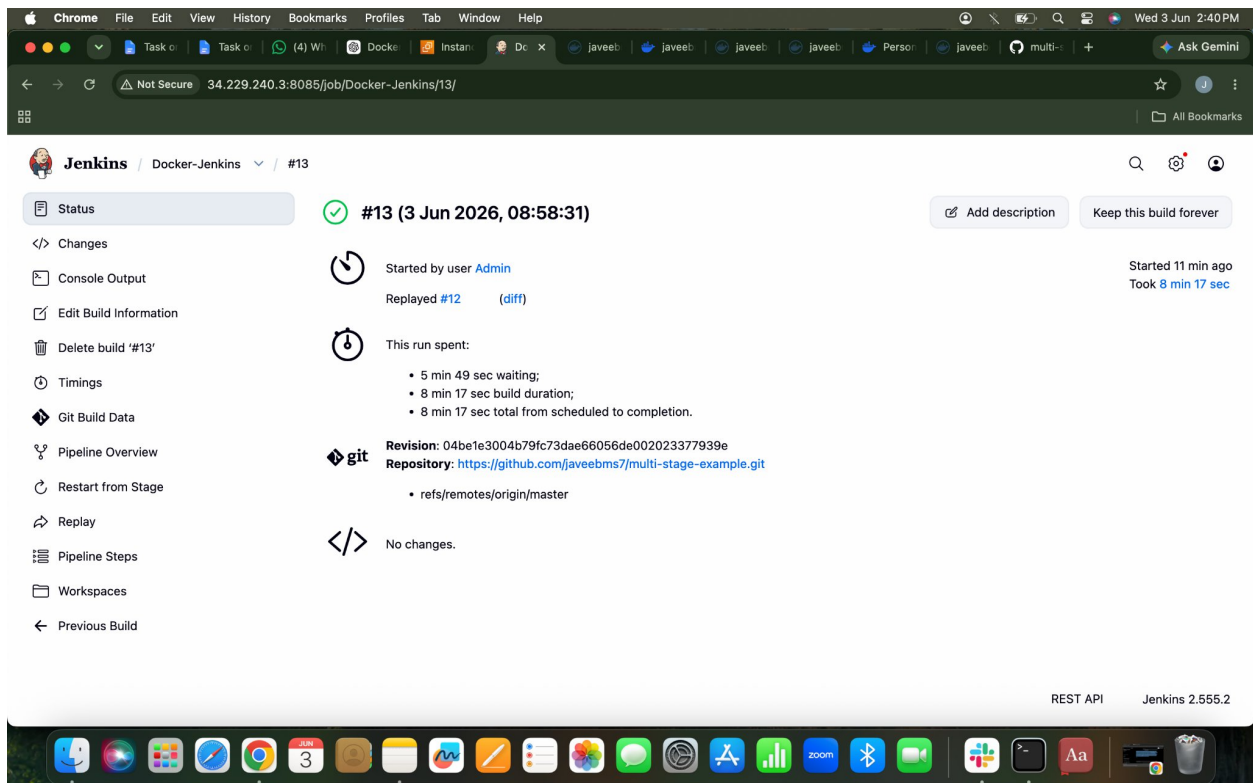
post {
    success {
        echo 'Docker image pushed successfully!'
    }

    failure {
        echo 'Pipeline failed!'
    }
}
}

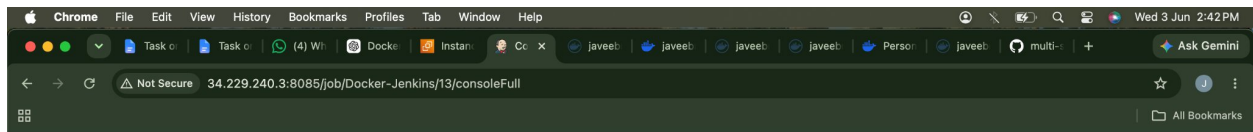
```



Build Successful



Console Output



Jenkins / Docker-Jenkins / #13

Status

Changes

Console Output

Edit Build Information

Delete build '#13'

Timings

Git Build Data

Pipeline Overview

Restart from Stage

Replay

Pipeline Steps

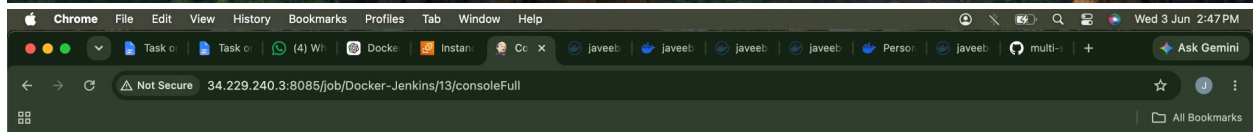
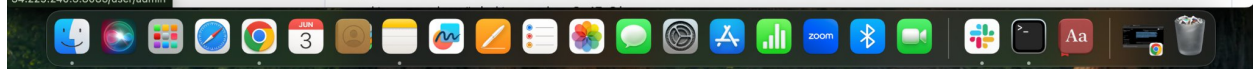
Workspaces

Previous Build

Console

Download Copy View as plain text

```
Started by user Admin
Replayed #12
[Pipeline] Start of Pipeline
[Pipeline] node
Still waiting to schedule task
Waiting for next available executor
Pausing (shutting down)
Resuming build at Wed Jun 03 09:04:26 UTC 2026 after Jenkins restart
Ready to run at Wed Jun 03 09:04:26 UTC 2026
Running on Jenkins in /var/jenkins_home/workspace/Docker-Jenkins
[Pipeline] {
[Pipeline] withEnv
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Clone Code)
[Pipeline] git
The recommended git tool is: NONE
No credentials specified
> git rev-parse --resolve-git-dir /var/jenkins_home/workspace/Docker-Jenkins/.git # timeout=10
Fetching changes from the remote Git repository
> git config remote.origin.url https://github.com/javeebms7/multi-stage-example.git # timeout=10
Fetching upstream changes from https://github.com/javeebms7/multi-stage-example.git
> git --version # timeout=10
```



Jenkins / Docker-Jenkins / #13

Status

Changes

Console Output

Edit Build Information

Delete build '#13'

Timings

Git Build Data

Pipeline Overview

Restart from Stage

Replay

Pipeline Steps

Workspaces

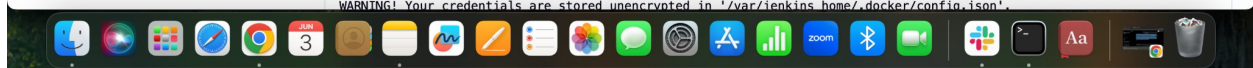
Previous Build

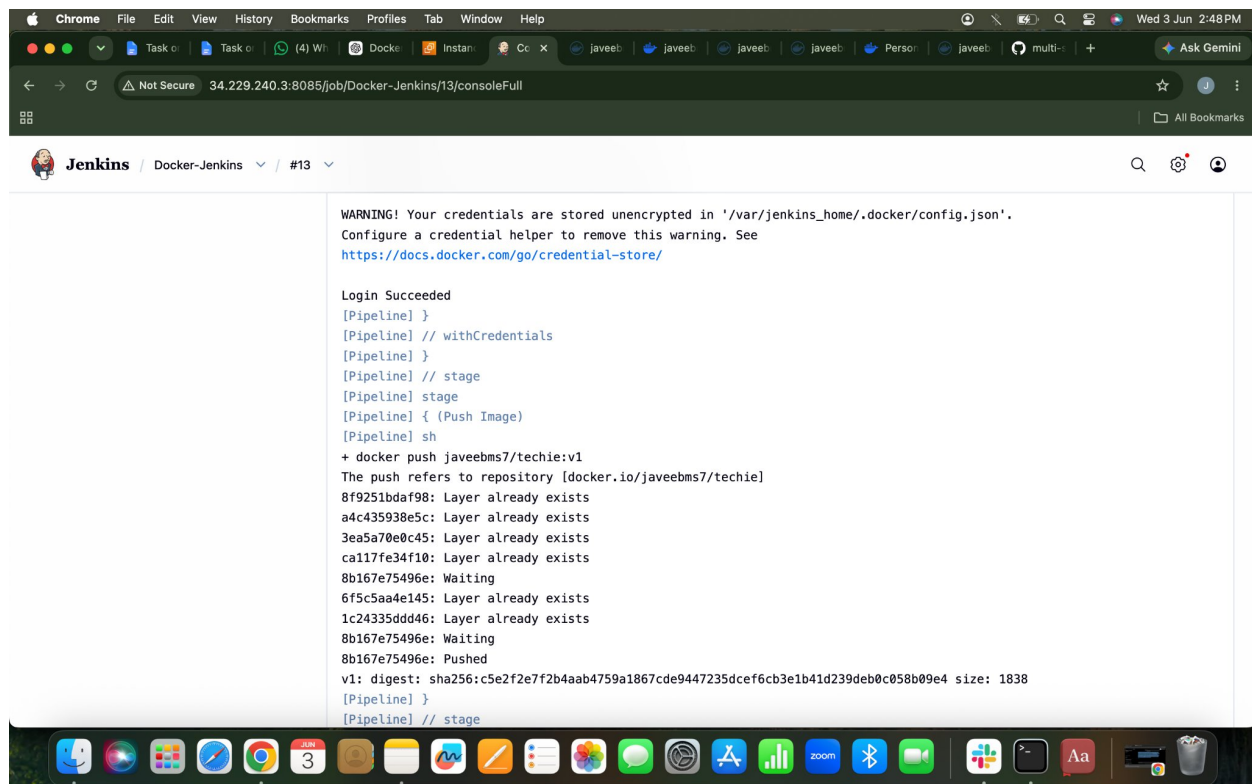
Console

Download Copy View as plain text

```
Digest: sha256:1ef686a85b239b143271093e286c27c6659051ba6b1c327e3a056742b13889a5
Status: Downloaded newer image for eclipse-temurin:8-jre
----> 1ef686a85b23
Step 7/9 : COPY --from=builder /app/source/target/*.jar /app/app.jar
----> 0109eff368c8
Step 8/9 : EXPOSE 8080
----> Running in bf87f183fdff
----> Removed intermediate container bf87f183fdff
----> 8cc5b48d1cf1
Step 9/9 : ENTRYPOINT ["java","-Djava.security.egd=file:/dev/./urandom","-jar","/app/app.jar"]
----> Running in 597ebd841bd
----> Removed intermediate container 597ebd841bd
----> c5e2f2e7f2b4
Successfully built c5e2f2e7f2b4
Successfully tagged javeebms7/techie:v1
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Login DockerHub)
[Pipeline] withCredentials
Masking supported pattern matches of $DOCKER_PASS
[Pipeline] {
[Pipeline] sh
+ echo ***
+ docker login -u javeebms7 --password-stdin

WARNING! Your credentials are stored unencrypted in '/var/jenkins_home/.docker/config.json'.
```





Now lets check in the Dockerhub , Image has successfully pushed , we can check in Image Managemnt in our repo

